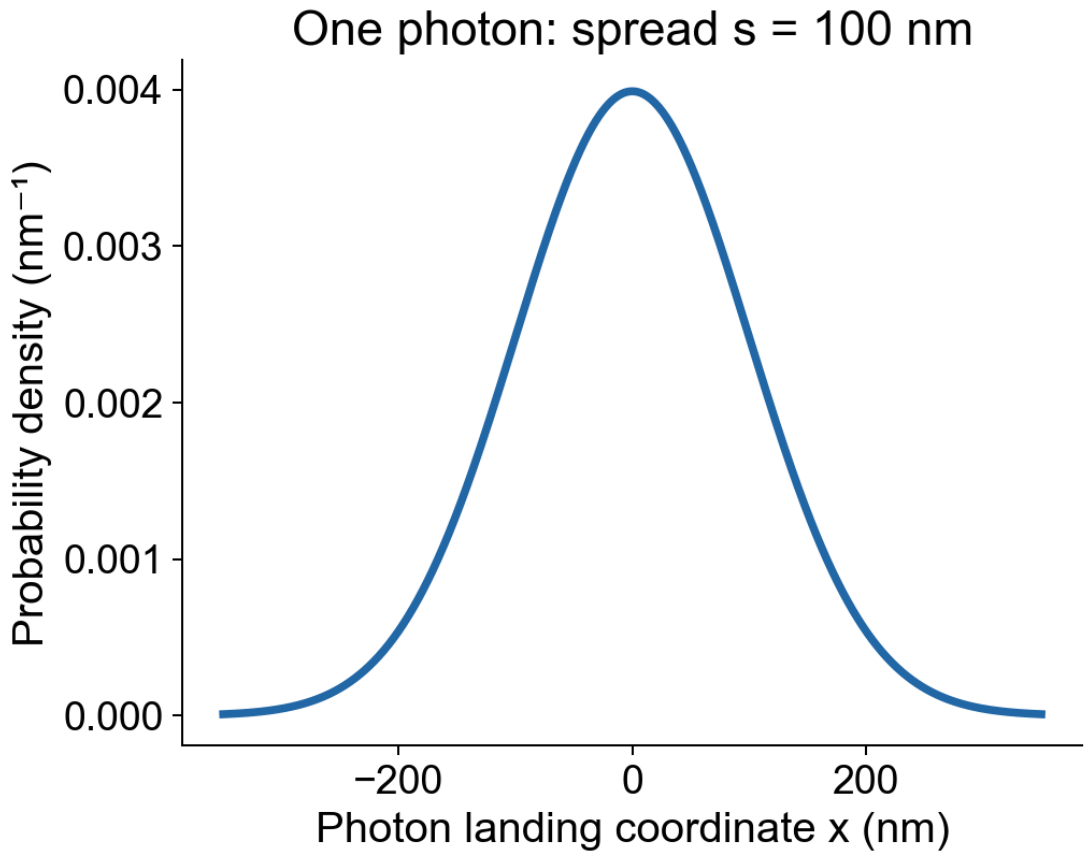
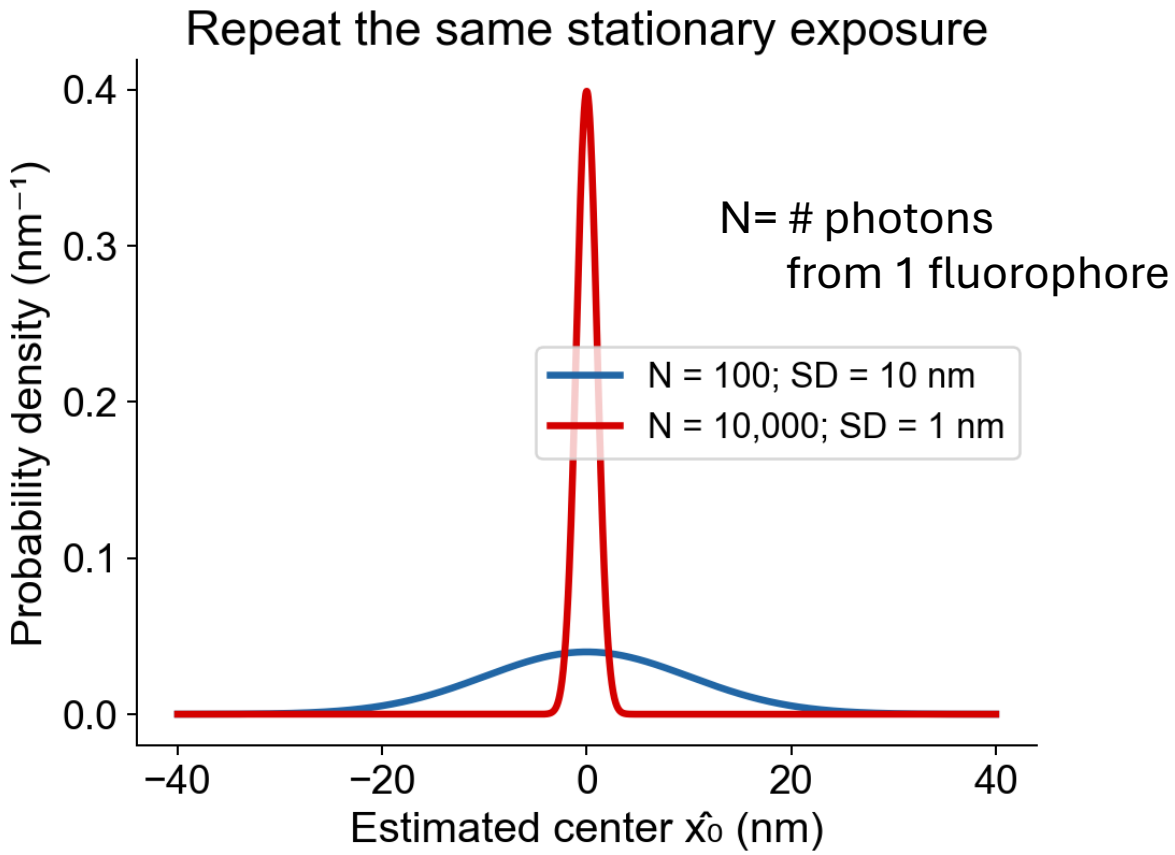
From a blurry spot to molecular structure

Physics of Living Systems · Lecture 8
October 22, 2026

Photon statistics, optical sectioning, STORM, STED, MINFLUX, and force

Start with one stationary molecule





The PSF spreads individual photons.
Repeated photons constrain the same center.

Ideal 1D Gaussian model; one isolated stationary emitter; no background. Both panels are normalized densities.

Define the unknown position and its estimate

Let x_0 be the true position. Photon i lands at x_i.

$$x_i = x_0 + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = s^2$$

$$\hat{x}_0 \equiv \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\mathbb{E}[\hat{x}_0] = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[x_i] = \frac{Nx_0}{N} = x_0$$

The hat means "estimated from these data." One exposure gives one value of $\hat{x}_0$.

s is the standard deviation of one photon's landing coordinate. N is fixed in this calculation.

Why does the variance decrease as 1/N?

Each photon adds a random error. Independent errors have zero covariance.

$$\hat{x}_0 - x_0 = \frac{1}{N}\sum_{i=1}^{N}\varepsilon_i$$

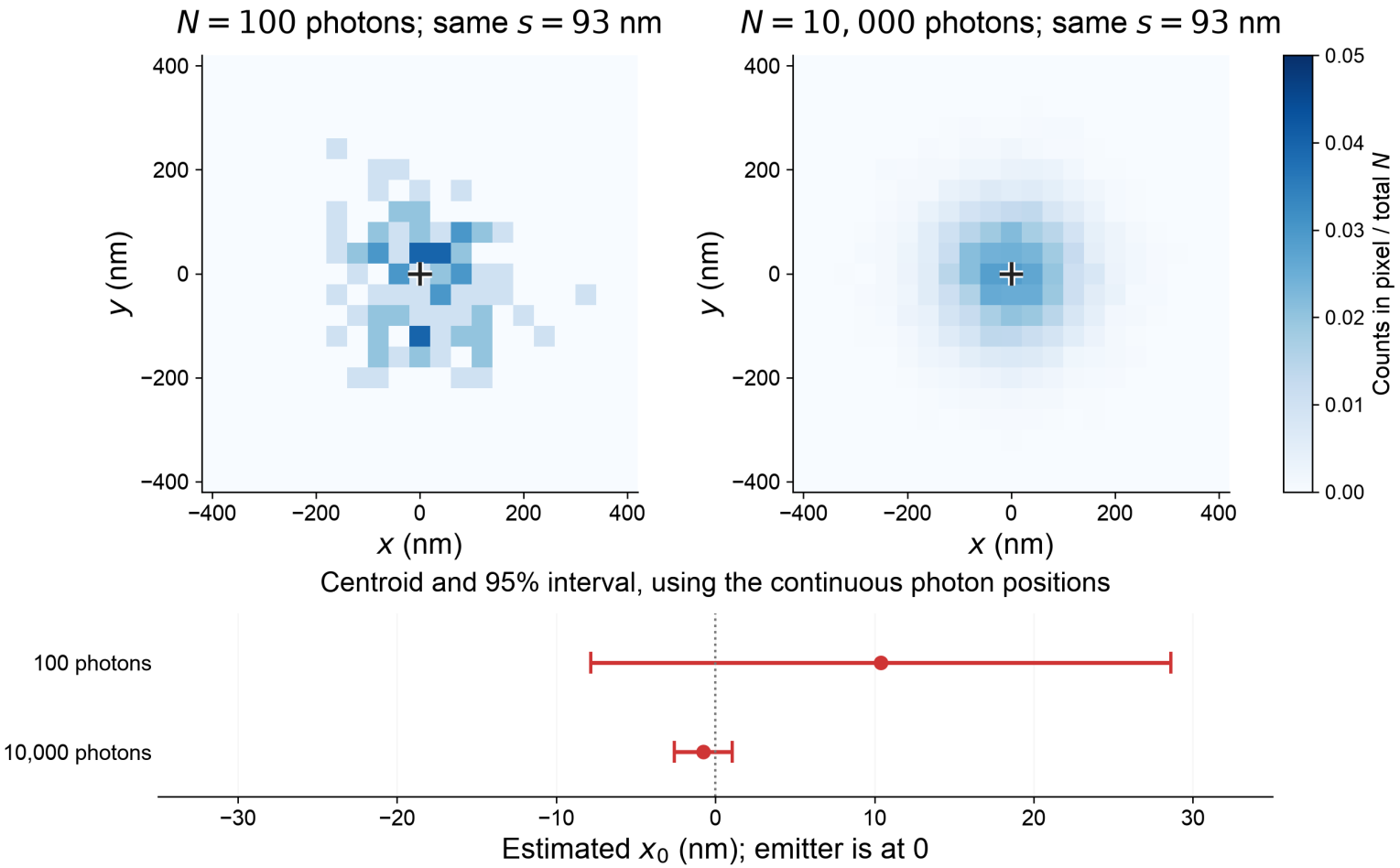
$$\mathrm{Var}(\hat{x}_0) = \frac{1}{N^2}\left[\sum_i \mathrm{Var}(\varepsilon_i) + 2\sum_{i<j}\mathrm{Cov}(\varepsilon_i, \varepsilon_j)\right]$$

$$= \frac{Ns^2}{N^2} = \frac{s^2}{N}, \qquad \sigma_{\mathrm{loc}} = \frac{s}{\sqrt{N}}$$

We divide a sum of N variances by N² because we average N measurements.

Independence is the key assumption. Drift and motion can correlate the errors or change the position being estimated.

More photons improve the center estimate



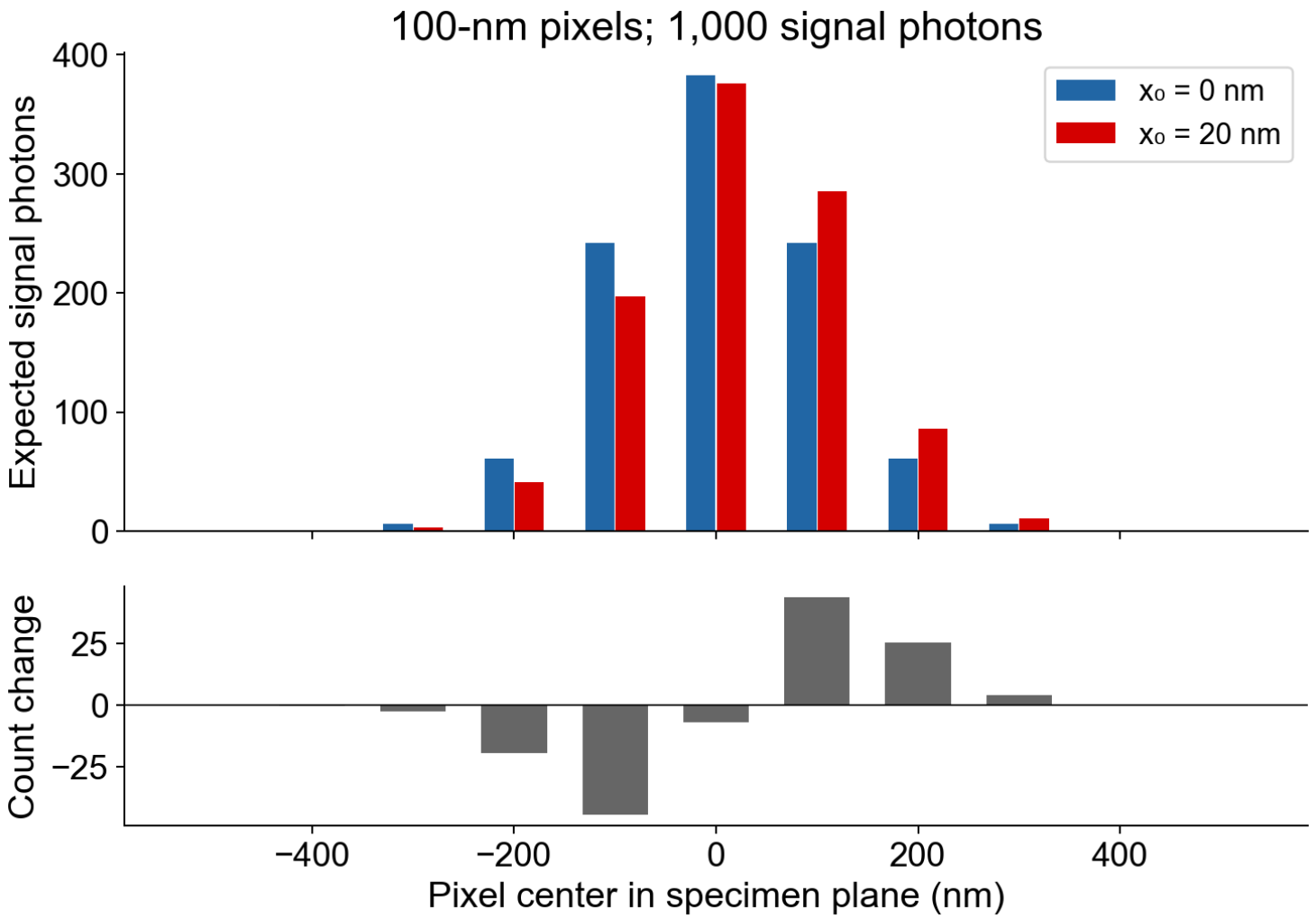
Same optics
Same stationary emitter
Same PSF width

100× more photons
⟶ 10× smaller SD

A longer exposure may average over motion.

Simulated Gaussian photons: s = 93 nm. Center intervals use continuous photon positions, not rounded pixel centers.

A subpixel shift changes several pixel counts



A camera reports counts in each pixel.

Moving a molecule redistributes photons among neighboring pixels.

The pattern changes continuously with position.

1D Gaussian example: s = 100 nm, a = 100 nm, N = 1,000. Expected counts; no simulated measurement noise.

How to detect motion smaller than pixel size

Integrate the PSF over a pixel before predicting its count.

Probability of a photon landing in pixel i: $$p_i(x_0) = \int_{\text{pixel } i} h(x - x_0)\, dx$$

Mean # of photons landing in pixel i: $$\mu_i(x_0) = N_s p_i(x_0) + B_i$$

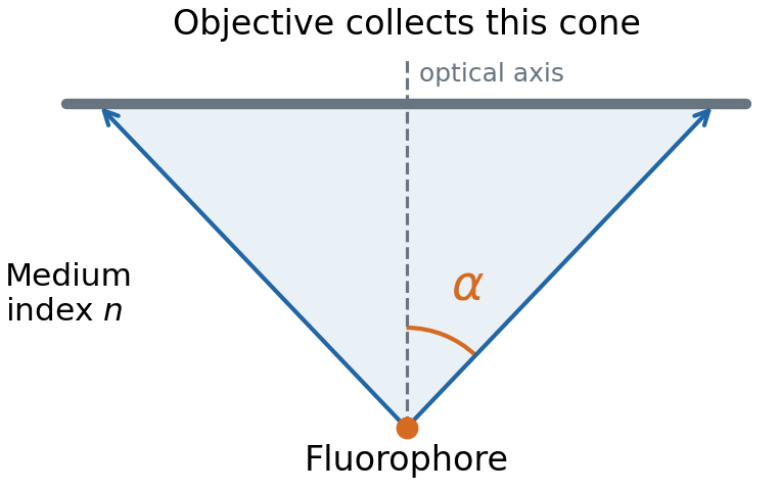
Change in mean # of photons in pixel i: $$\Delta\mu_i \simeq N_s \frac{dp_i}{dx_0} \Delta x_0$$

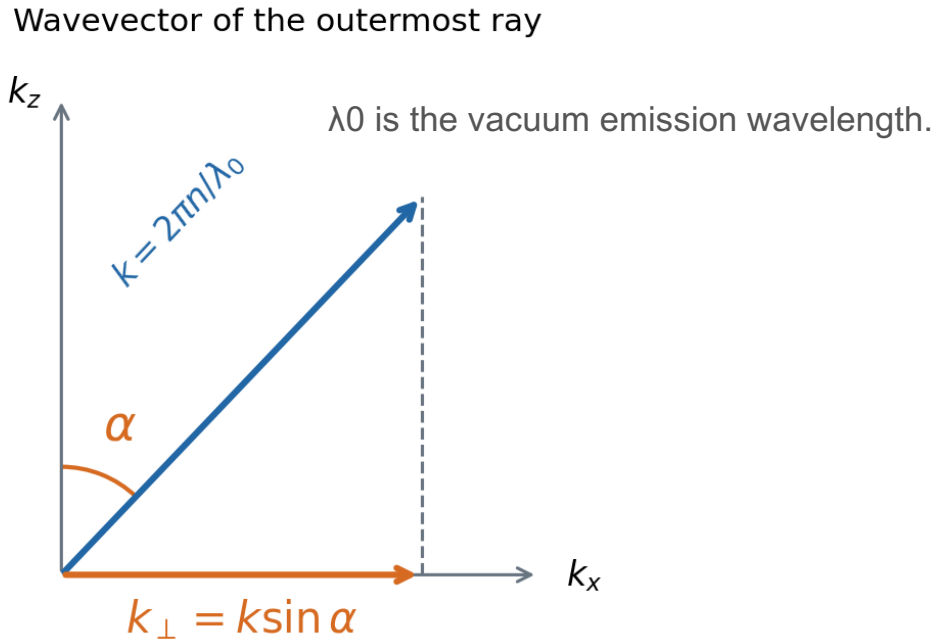
We infer shifts from count changes.
A shift can be much smaller than the pixel width.

N_s is expected signal count; B_i is mean background count. The 2D version integrates over pixel area.

Numerical aperture (NA): the collection cone

n is the refractive index. α is the half-angle from the optical axis to the outermost collected ray.





$$\mathrm{NA} = n\sin\alpha$$

Numerical aperture combines the collection angle with the refractive index.

Define n and α in the same medium. We begin with a homogeneous-medium model.

The refractive index sets the wavelength in the medium

The frequency v stays fixed at a stationary interface. The speed becomes c/n.

$$\nu = \frac{c}{n}, \qquad \lambda_{\text{med}} = \frac{v}{\nu} = \frac{c/n}{\nu} = \frac{\lambda_0}{n}$$

$$k_0 = \frac{2\pi}{\lambda_0}$$

$$k = \frac{2\pi}{\lambda_{\text{med}}} = \frac{2\pi n}{\lambda_0} = nk_0$$

A higher refractive index gives a shorter wavelength and a larger wavevector.

λ0 is the vacuum emission wavelength. k0 is the vacuum wavenumber. k is the wavenumber in the medium.

The sine comes from the transverse wavevector

A ray at angle θ to the optical axis has an axial and a transverse component.

$$k_\perp = k\sin\theta, \qquad k_z = k\cos\theta$$

$$K \equiv \max|\mathbf{k}_\perp| = k\sin\alpha = nk_0\sin\alpha$$

$$\mathrm{NA} \equiv \frac{K}{k_0} = n\sin\alpha$$

NA is the largest collected transverse wavevector, measured in units of k0.

The objective accepts θ ≤ α. K is the maximum magnitude of the transverse wavevector.

Transverse wavevectors determine the PSF width

At fixed z and t, the phase varies across the specimen as kx x.

$$E(x, z, t) \propto e^{i(k_x x + k_z z - \omega t)}$$

$$\Delta\phi = k_x \Delta x, \qquad \ell_x = \frac{2\pi}{|k_x|}$$

The lens combines these directions, as in a wave packet.
A circular aperture gives the Airy first-zero radius:

$$r_0 \simeq 0.61 \frac{\lambda_0}{\mathrm{NA}}$$

A broader range of transverse wavevectors allows a narrower point-source image.

r0 is the first-zero radius for an ideal circular scalar pupil. It is not a pixel size or localization uncertainty.

Why immersion oil can increase NA

Compare the same half-angle $\alpha = 60°$ at vacuum emission wavelength $\lambda_0 = 600$ nm.

Medium	n	NA = n sin 60°	First-zero radius
Air	1.00	0.866	423 nm
Oil	1.50	1.299	282 nm

In air, NA ≤ 1. A medium with n > 1 permits NA > 1.

Why NA matters for FIONA

A larger NA narrows the optical PSF, reducing the width s of its Gaussian approximation.

A larger collection angle can also increase the number N of detected signal photons.

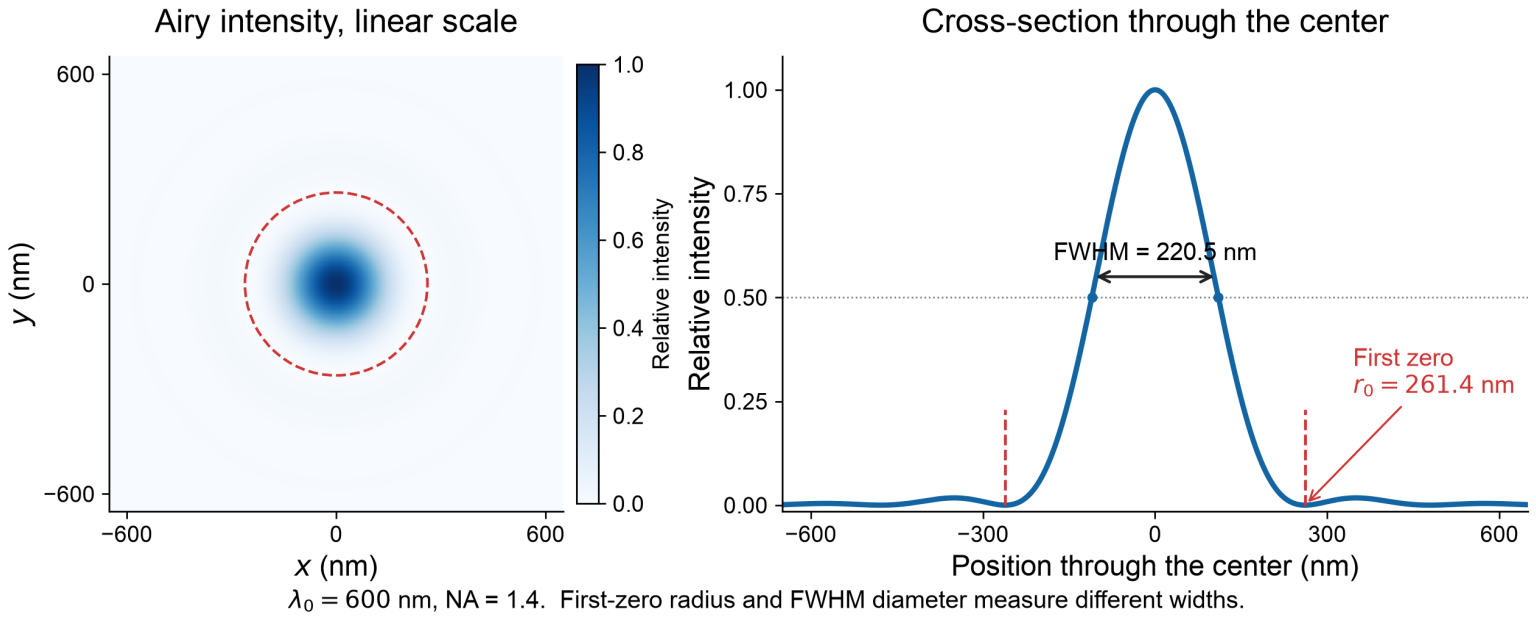
$$\sigma_{\mathrm{loc}} \simeq \frac{s}{\sqrt{N}}$$

s: Gaussian PSF width N: detected signal photons

A narrower spot and more detected photons can both improve localization precision.

Ideal isolated stationary emitter. Background, aberrations, drift and label motion also affect precision.

Diffraction spreads the image of a point source



λ_0 = 600 nm, NA = 1.4. First-zero radius and FWHM diameter measure different widths.

The objective collects only a finite cone of light.

NA = n sin α

A finite aperture spreads the image of one emitter.

Two equal sources: Rayleigh separation
d ≈ 0.61 λ / NA

Circular-pupil Airy model: λ = 600 nm, NA = 1.4. Rayleigh separation ≈ 261 nm; one-emitter localization can be much finer.

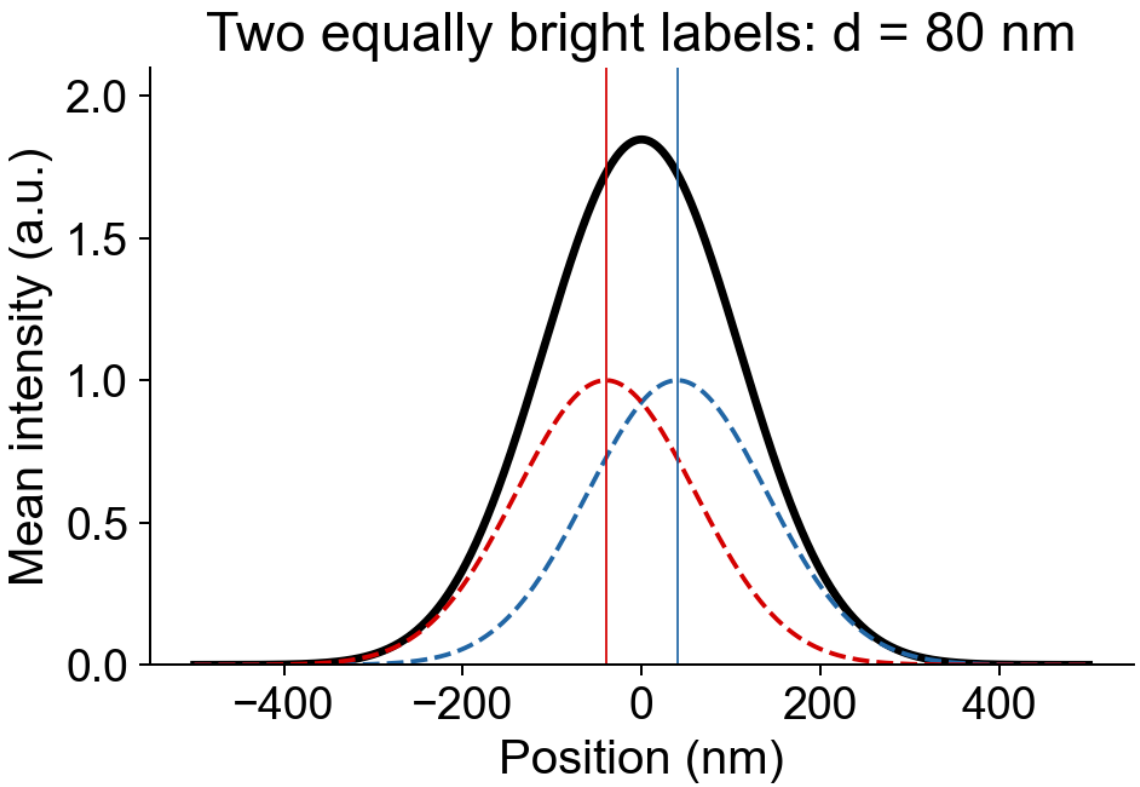
Three lengths answer three different questions

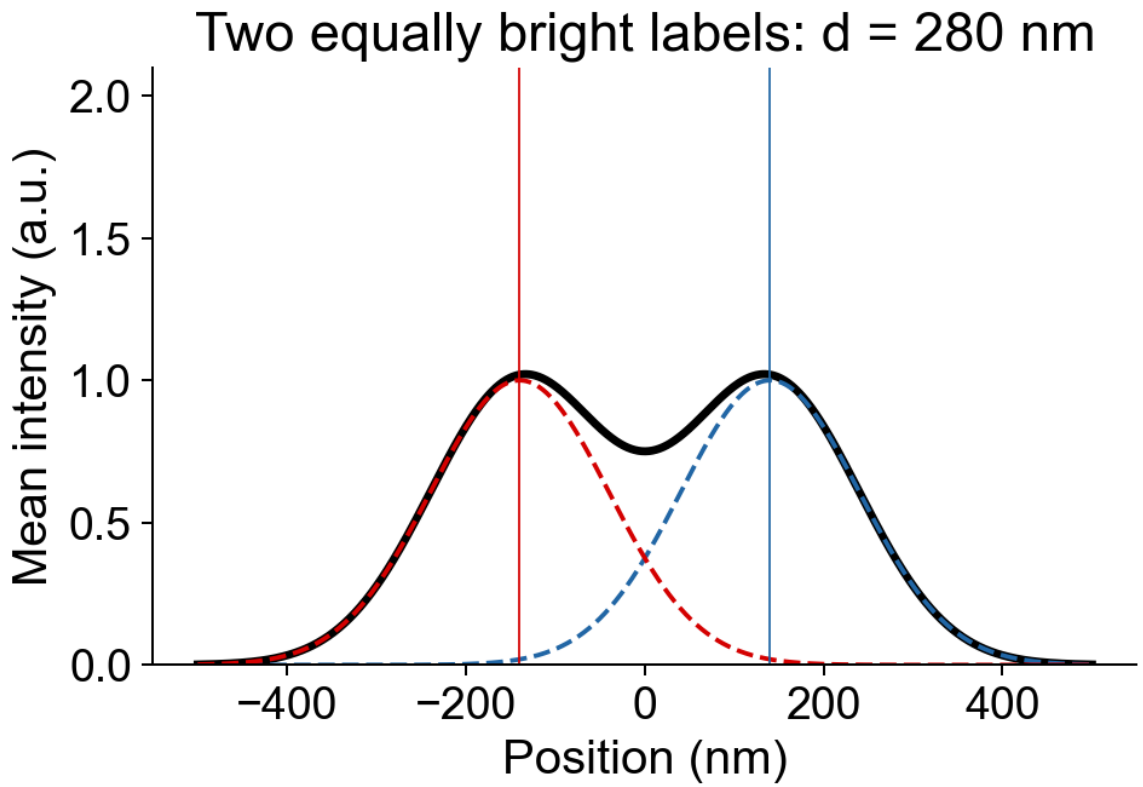
Quantity	Physical meaning	Illustrative scale
Optical resolution	Can nearby sources be distinguished?	~260 nm
Pixel width a	How does the camera sample the image?	65 nm
Localization SD	How uncertain is one fitted center?	3.2 nm

Example: λ = 600 nm, NA = 1.4; 6.5-μm sensor pixels at 100×; s = 100 nm, N = 1,000.

Optical scale: Rayleigh separation 0.61 λ/NA ≈ 260 nm. Localization example: ideal s/√N.

One precise center does not separate two molecules





With two emitters, we must infer two positions and assign their overlapping signals.

Illustration with Gaussian PSFs (s = 100 nm). The separation problem differs from fitting one known isolated emitter.

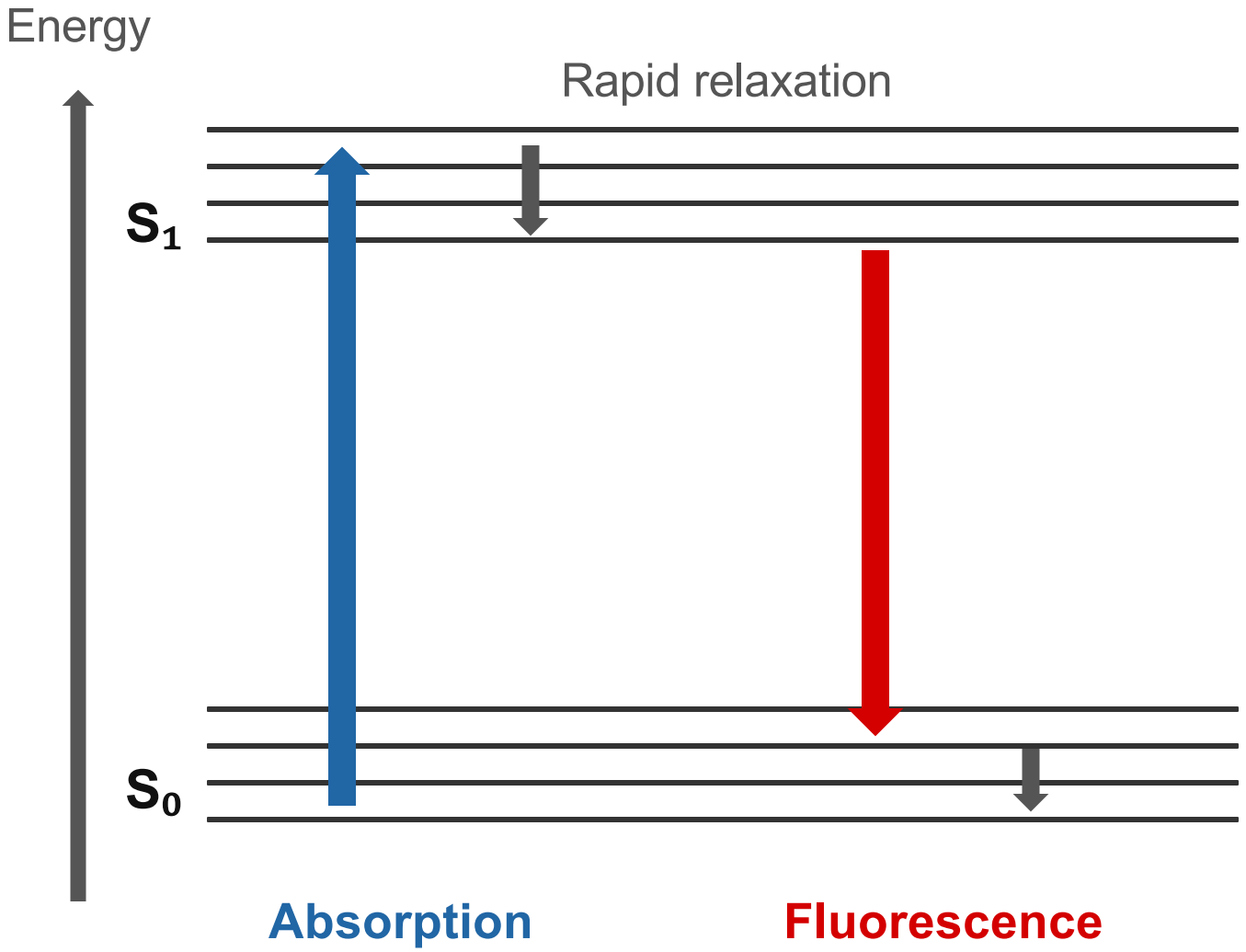
Control which molecules contribute photons

Strategy	What we control	Method
Separate emitters in time	A sparse set is bright in each frame	STORM / PALM
Restrict emission in space	Deplete fluorescence around a zero	STED
Probe an isolated emitter	Move a known excitation minimum	MINFLUX

First consider spatial selection: confocal detection and light-sheet excitation reduce background.

STORM and STED established the principles; MINFLUX is a newer development built around a controlled intensity minimum.

Fluorescence cycles through molecular states

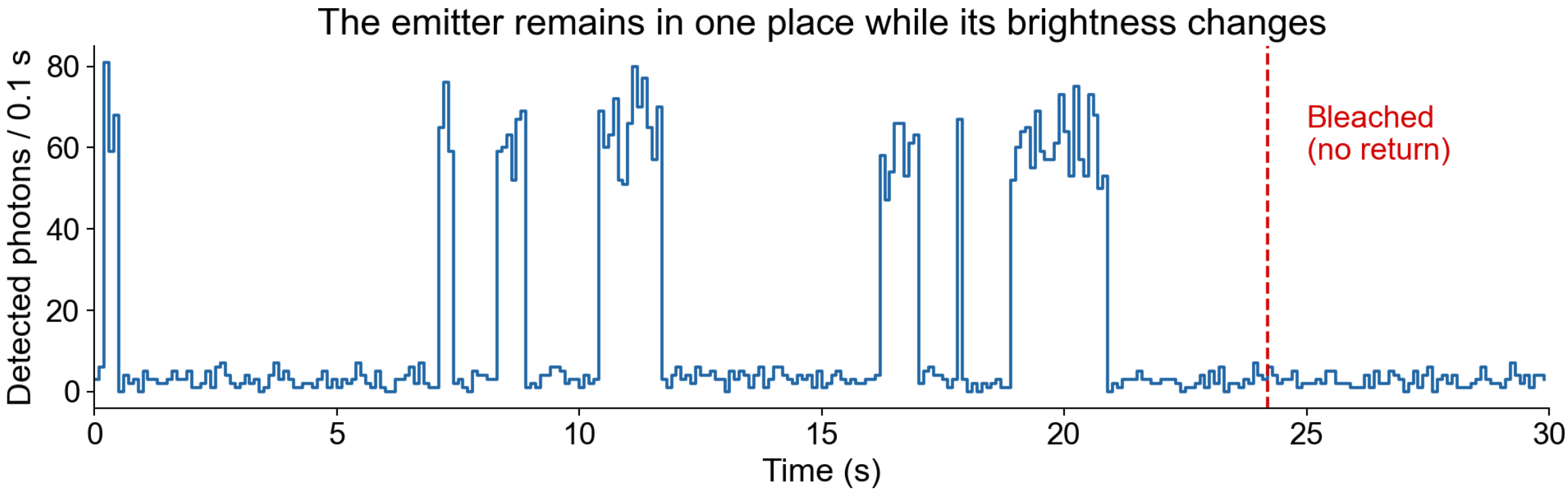


Absorb a photon.
Relax within S_1.
Emit a lower-energy photon.
Return to S_0.

Vibration changes nuclear positions around the equilibrium geometry.

S_0 and S_1 are electronic singlet states. Horizontal sublevels represent nuclear vibrational excitation.

Blinking changes brightness while the molecule stays put



Blinking: the molecule returns from a dark state. Bleaching: it loses fluorescence irreversibly.

Simulated two-state trajectory with Poisson photon counts and one imposed bleaching event.

Different physical mechanisms can create a dark state

Dark state	Why fluorescence stops	How brightness returns
Triplet electronic state	Spin change makes return to S_0 slow	Triplet decay or quenching
Radical / chemical adduct	Electronic structure or absorption changes	Reverse redox / chemical reaction
Different conformation or protonation	Excitation or emission pathway changes	Thermal or light-driven conversion

“OFF” describes an observation. It does not identify a unique molecular state.

Mechanisms depend on the dye, solvent, oxygen, illumination and buffer.

For cyanine STORM, light and buffer control a switch

Cy5 can form a long-lived dark state by reversibly attaching a chemical group.

A thiol is a molecule with an –SH group.

Light can make it attach to Cy5's light-absorbing electronic system.

This adduct (dye + attached group) changes absorption and becomes dark.

Shorter-wavelength light can restore fluorescent Cy5.

We use photochemistry to make only a small fraction of labels bright at one time.

A rate model predicts the fraction of bright molecules

Let k_on take OFF → ON and k_off take ON → OFF.

$$\frac{dp_{\mathrm{on}}}{dt} = k_{\mathrm{on}}(1 - p_{\mathrm{on}}) - k_{\mathrm{off}} p_{\mathrm{on}}$$

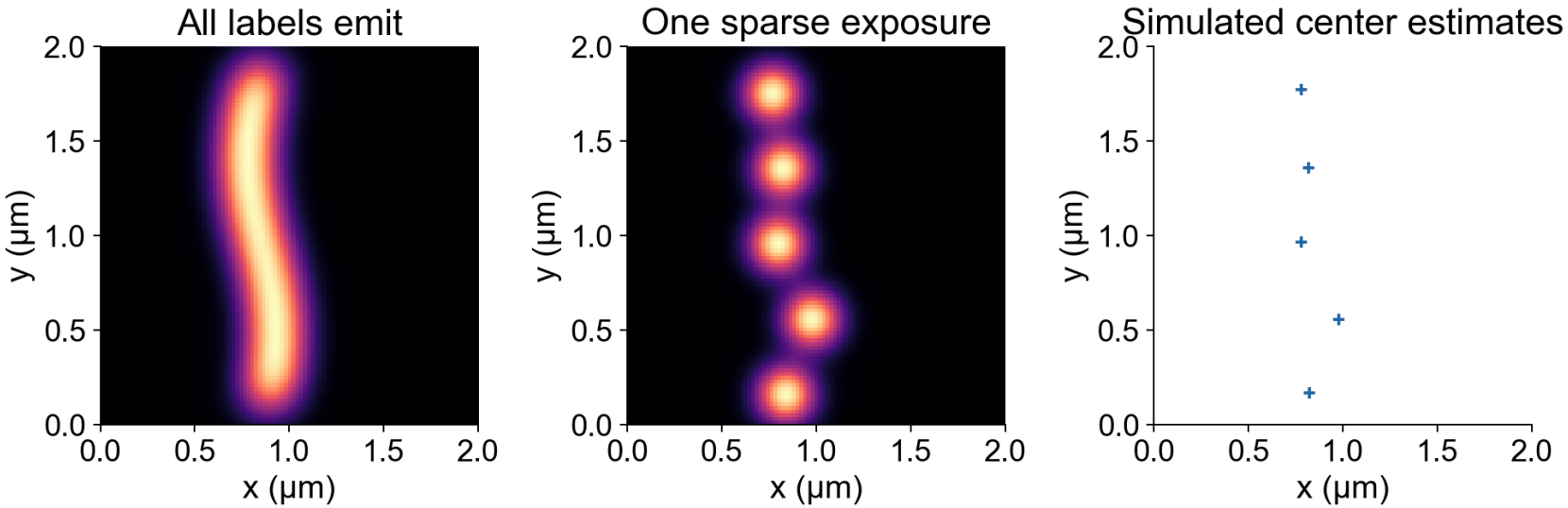
$$0 = k_{\mathrm{on}} - (k_{\mathrm{on}} + k_{\mathrm{off}}) p_{\mathrm{on}}^{*}$$

$$p_{\mathrm{on}}^{*} = \frac{k_{\mathrm{on}}}{k_{\mathrm{on}} + k_{\mathrm{off}}} = \frac{\tau_{\mathrm{on}}}{\tau_{\mathrm{on}} + \tau_{\mathrm{off}}}$$

Long OFF periods and short bright bursts keep the duty cycle small.

Two-state Markov approximation: τ_on = 1/k_off and τ_off = 1/k_on. Real dyes can have several dark states.

STORM reconstructs a structure from sparse frames



1. Switch most labels OFF.
2. Read a sparse subset.
3. Fit each isolated PSF.
4. Store position and localization uncertainty.
5. Repeat with new subsets.

How sparse must the bright subset be?

Suppose M labels share one diffraction-limited region. Each is ON with probability p.

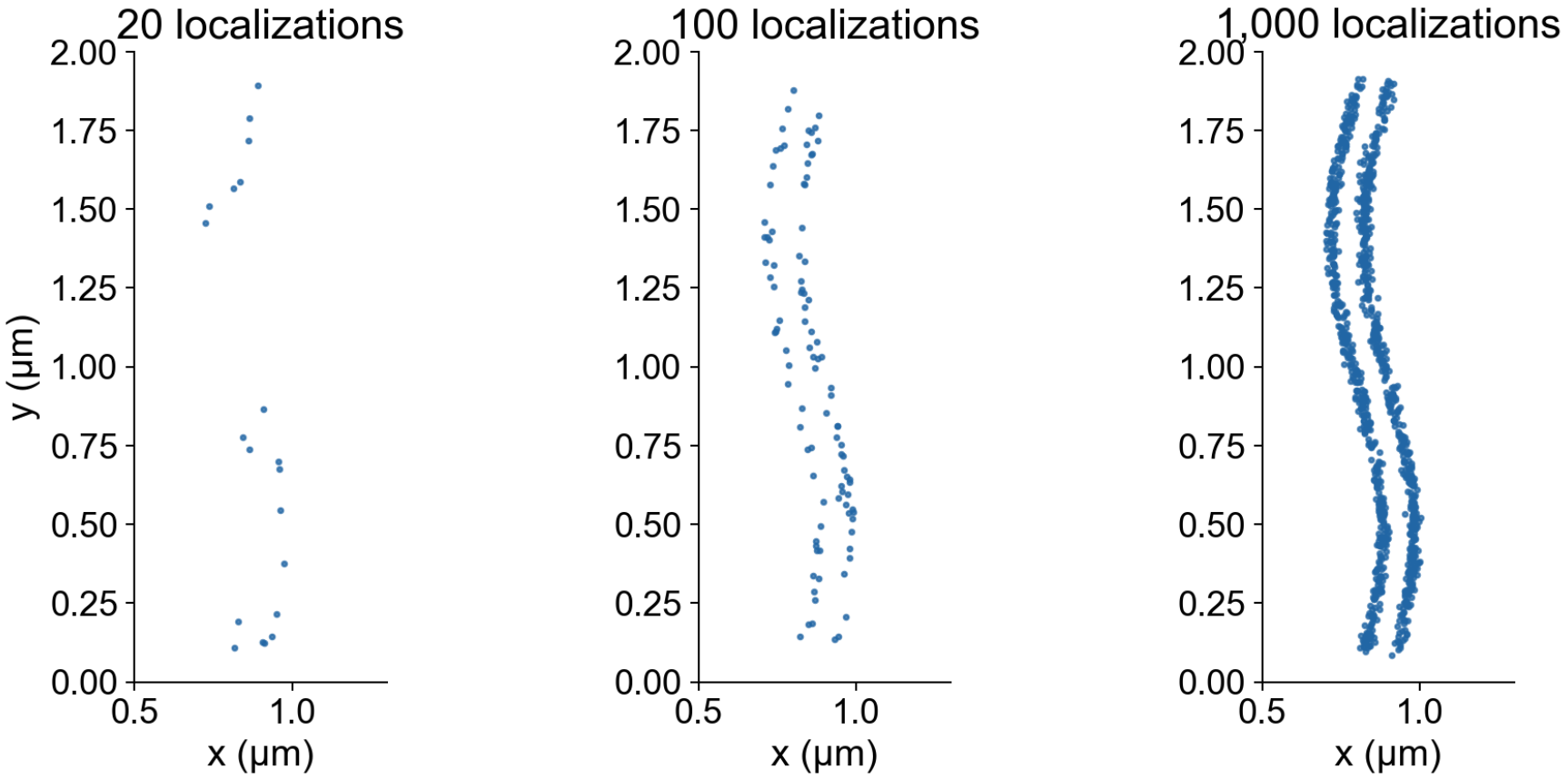
$$\rho = Mp, \qquad K \approx \text{Poisson}(\rho)$$

$$P(K \geq 2) = 1 - P(K = 0) - P(K = 1) = 1 - e^{-\rho}(1 + \rho)$$

$$P(K \geq 2 \mid K \geq 1) = \frac{1 - e^{-\rho}(1 + \rho)}{1 - e^{-\rho}} \simeq \frac{\rho}{2}$$

Example: M = 100 and p = 0.001 give ρ = 0.1
and ~4.9% overlap among nonempty frames.

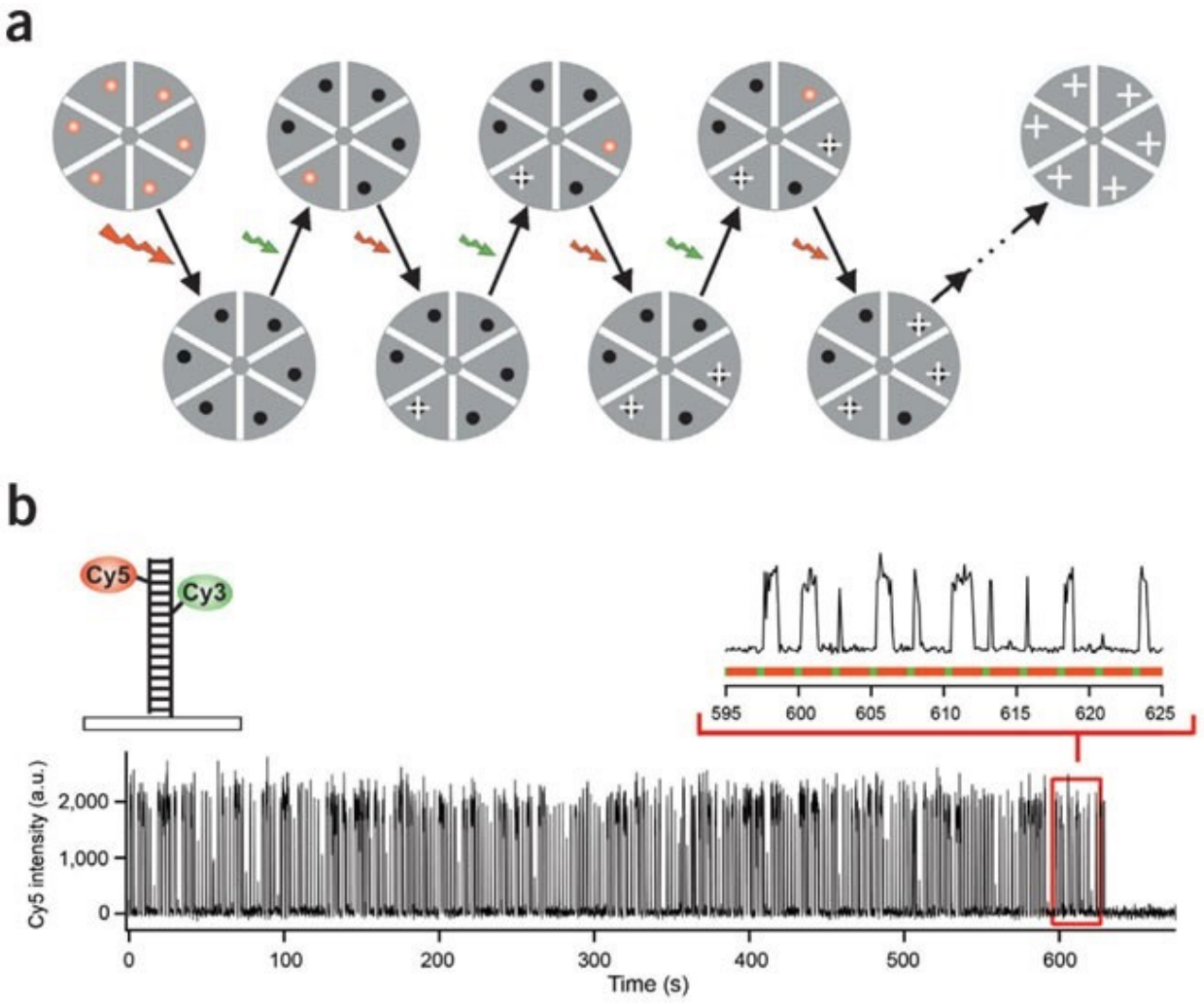
Repeated localizations fill in the structure



Localization precision sets the uncertainty of each point. Label coverage sets which parts we sample.

Simulation: two filaments 100 nm apart; independent 10-nm Gaussian localization errors. Repeated events can come from the same label.

The original STORM experiment switched and localized dyes



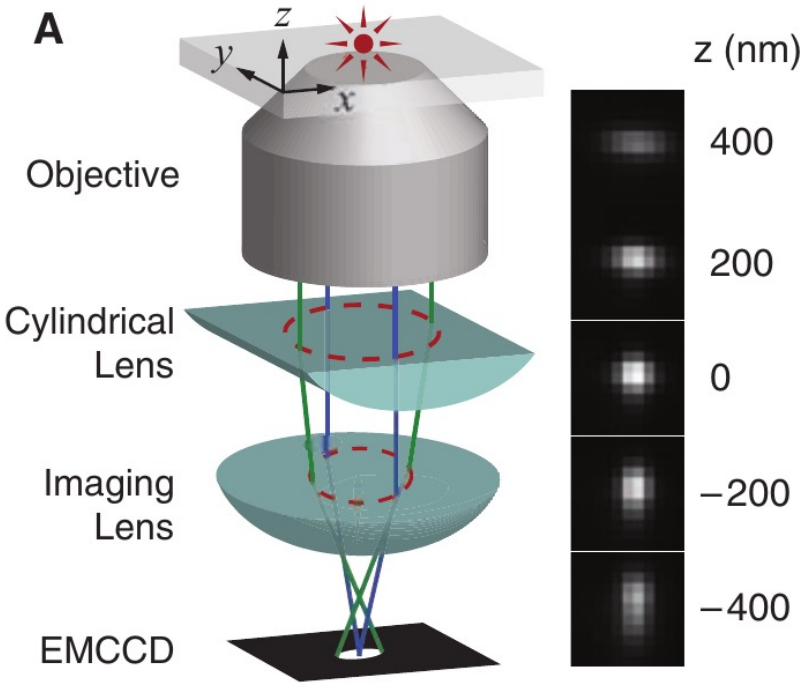
Cy3–Cy5 pairs acted as activator–reporter labels.

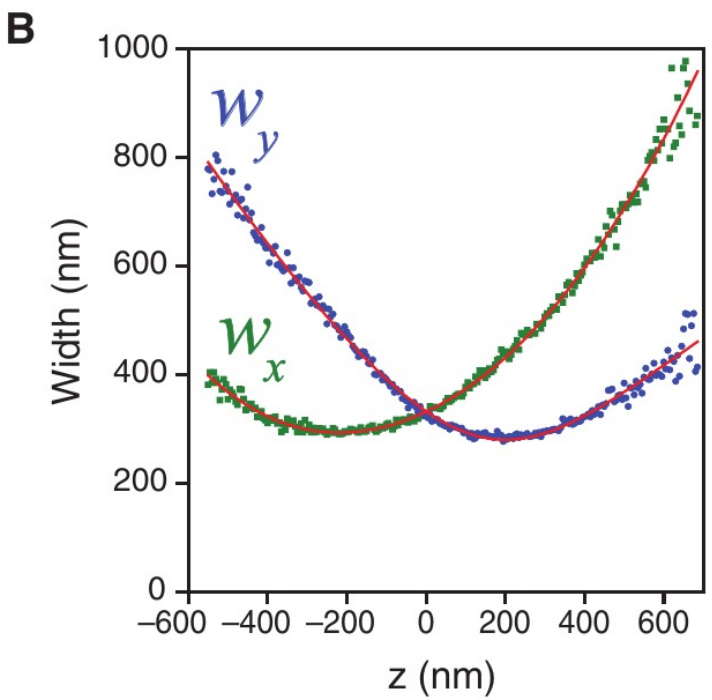
Reversible switching separated nearby emitters in different readout cycles.

Fit a position in each bright interval.

Rust, Bates & Zhuang, Nature Methods (2006). Original switching and localization experiment.

Astigmatism turns axial position into spot shape





A cylindrical lens gives x and y different focal planes.

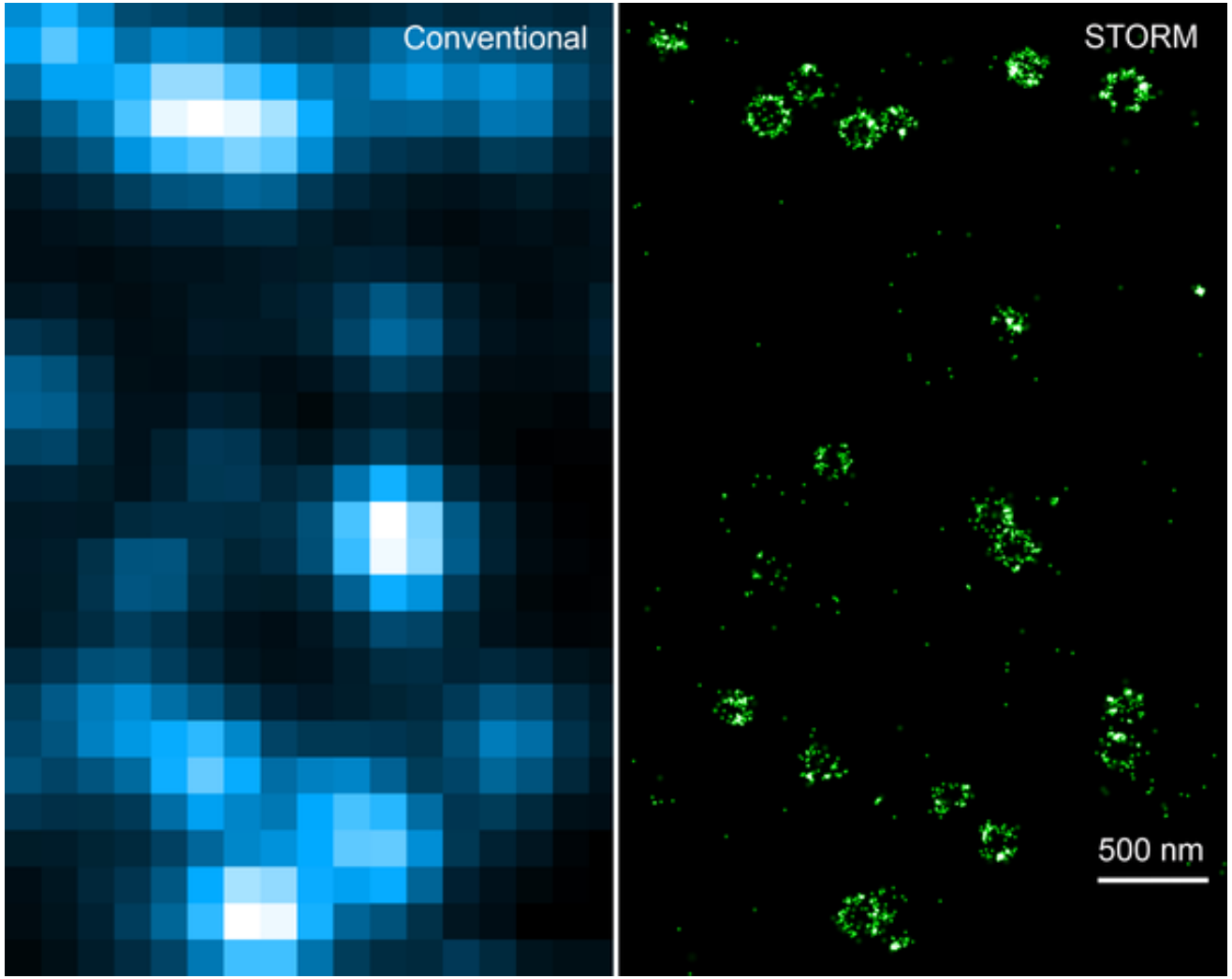
Calibrate both widths at known heights.

The measured width pair then reports z.

The microscope encodes height in a calibrated shape change.

Huang, Wang, Bates & Zhuang, Science (2008). Axial inference also requires photons and a suitable PSF model.

3D STORM reveals the shape of clathrin-coated pits



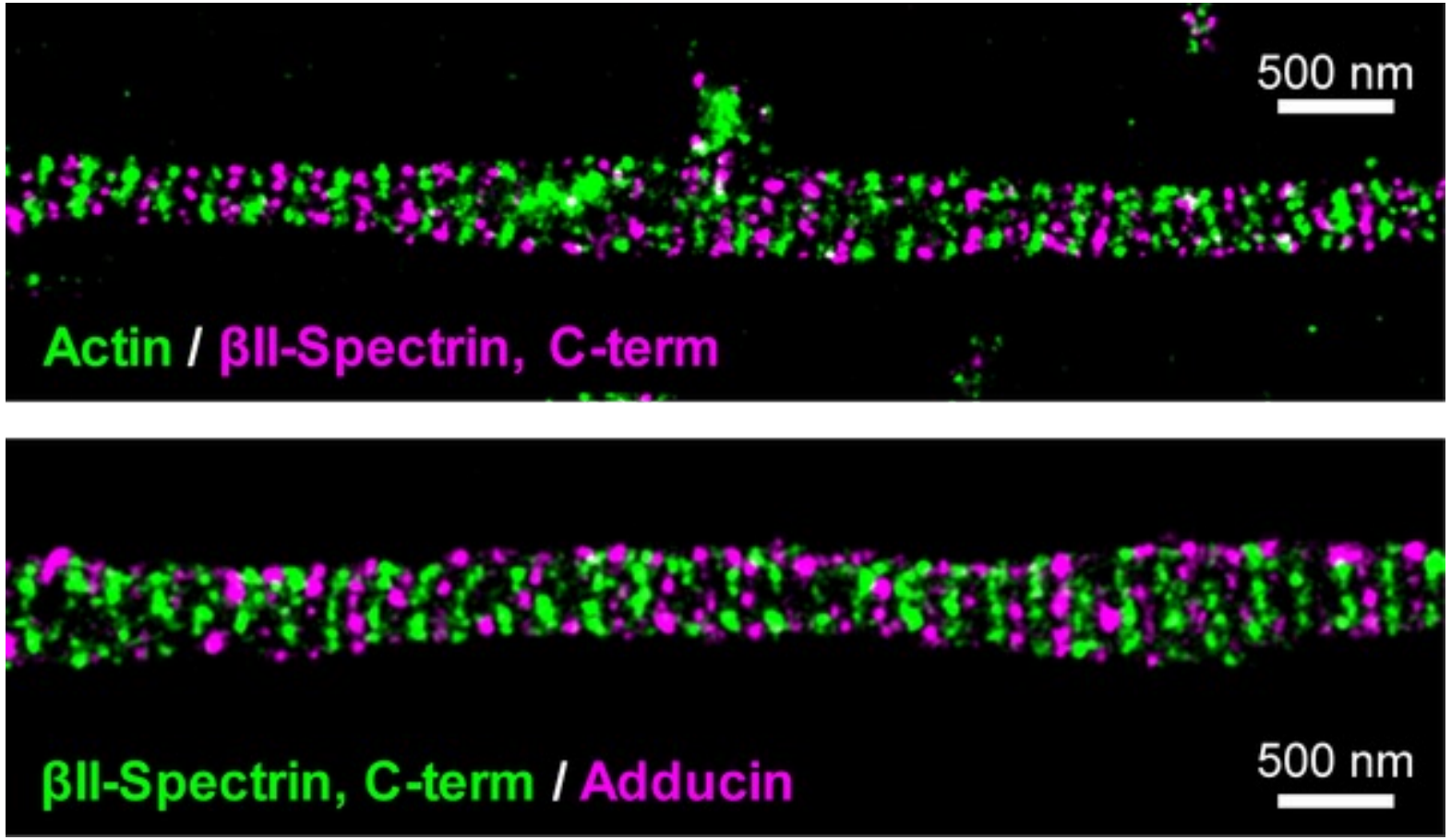
Clathrin forms coats around membrane buds.

A conventional image merges nearby structures.

This xy section separates clathrin structures in a 3D localization data set.

Huang et al., Science (2008). Conventional image and xy section of 3D STORM data; bar 500 nm.

STORM revealed a periodic skeleton in axons



An axon is a neuron's long signaling extension.

Actin forms circumferential rings near its membrane.

Spectrin connects adjacent rings: spacing ~190 nm.

Xu, Zhong & Zhuang, Science (2013). Fluorescent labels report the locations of specific cytoskeletal proteins.

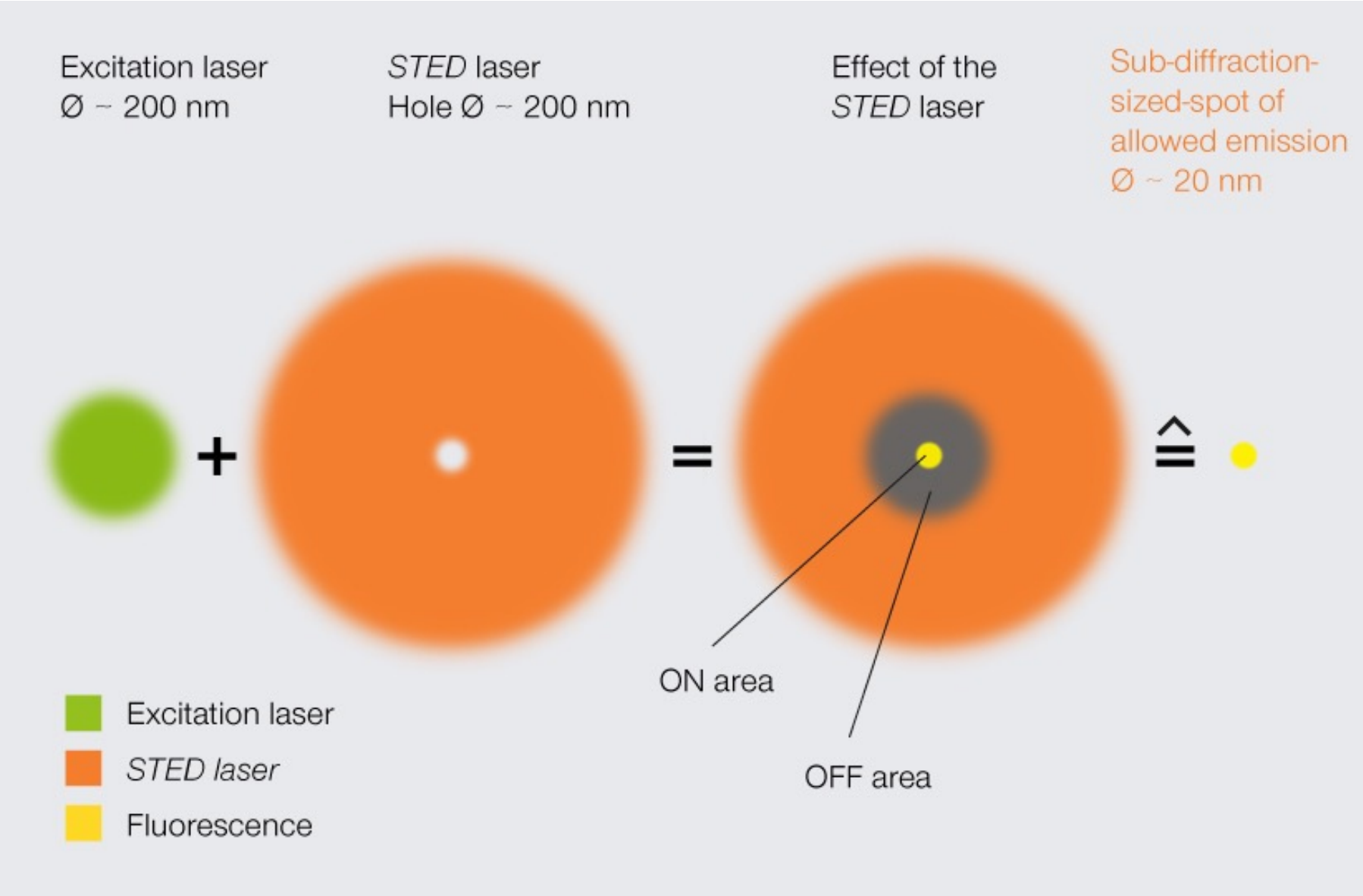
Sparse localization can use different molecular switches

Method	How labels become visible	What we repeat
STORM / dSTORM	Organic dyes switch between bright and dark states	Reversible switching
PALM	Often activate or convert fluorescent proteins	Activation, readout, bleaching
DNA-PAINT	Fluorescent DNA strands bind targets transiently	Binding and unbinding

Shared idea: collect identifiable emission events from a densely labeled structure.

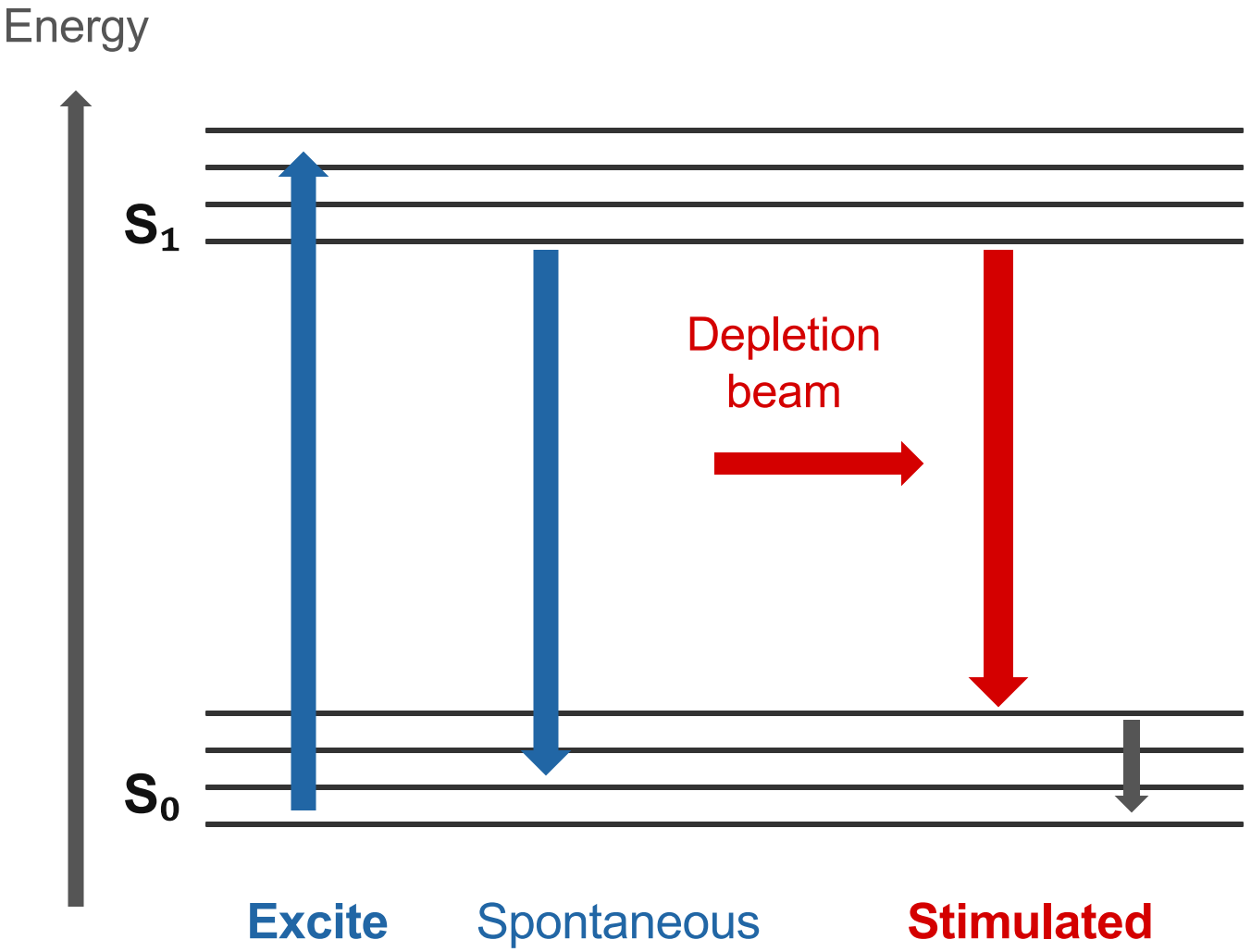
The method names describe typical implementations; switching labels and analysis strategies overlap.

STED = STimulated Emission Depletion



from abberior: https://abberior.rocks/knowledge-base/what-is-the-resolution-of-a-sted-microscope/

STED adds a competing path out of the excited state



A red-shifted beam drives stimulated emission.

The molecule returns to S_0 before it fluoresces spontaneously.

Filters reject the strong depletion light.

STED = stimulated emission depletion. The depletion wavelength lies in a suitable part of the emission spectrum.

Two ways for an excited molecule to emit

One excitation supplies energy for one electronic decay.

Spontaneous emission

Excited molecule → Ground-state molecule + 1 photon

The emitted photon can fall within the fluorescence detection band.

Stimulated emission

Excited molecule
+ m photons in the laser mode

→

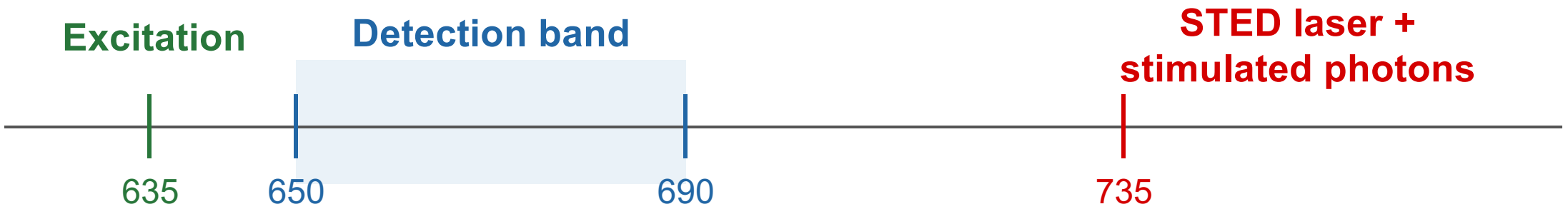
Ground-state molecule
+ (m + 1) photons in that mode

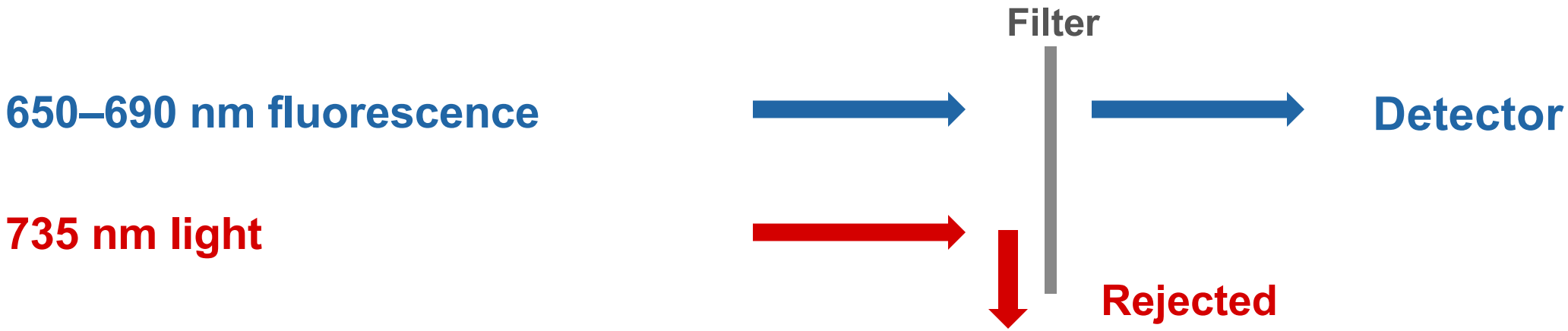
After this decay, the molecule must be excited again before it can fluoresce.

A mode specifies frequency, polarization and spatial field pattern. The molecule supplies the added photon's energy.

The detector selects a fluorescence wavelength band

The added stimulated photon has the depletion laser's wavelength.

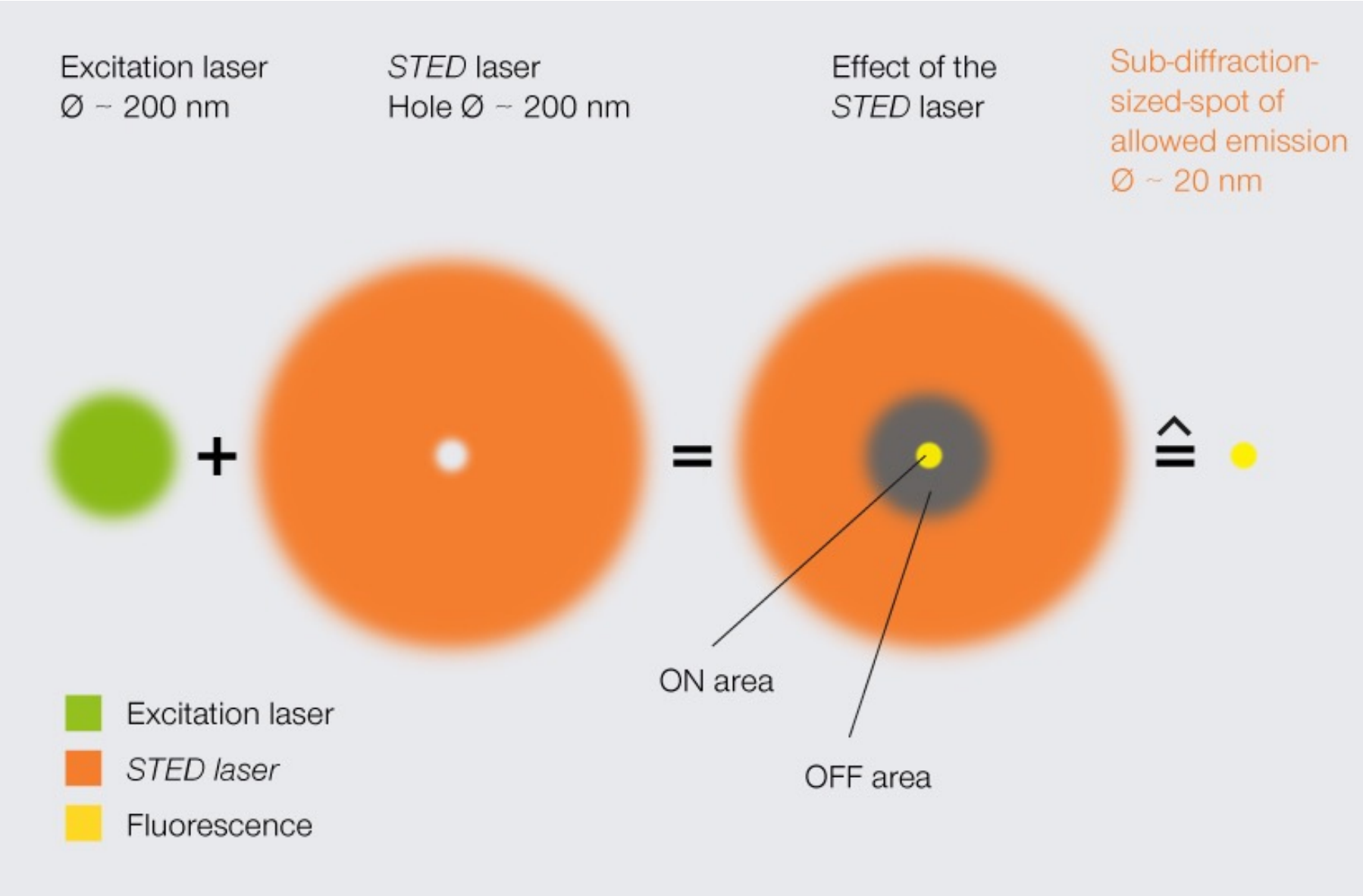




The same filter is present with STED off and on.
Turning STED on reduces spontaneous photons before they reach it.

Example wavelengths: Harke et al., Optics Express (2008), §3. Schematic filters; this example used pulsed STED.

STED = STimulated Emission Depletion



from abberior: https://abberior.rocks/knowledge-base/what-is-the-resolution-of-a-sted-microscope/

Photon accounting for 100 excitation events

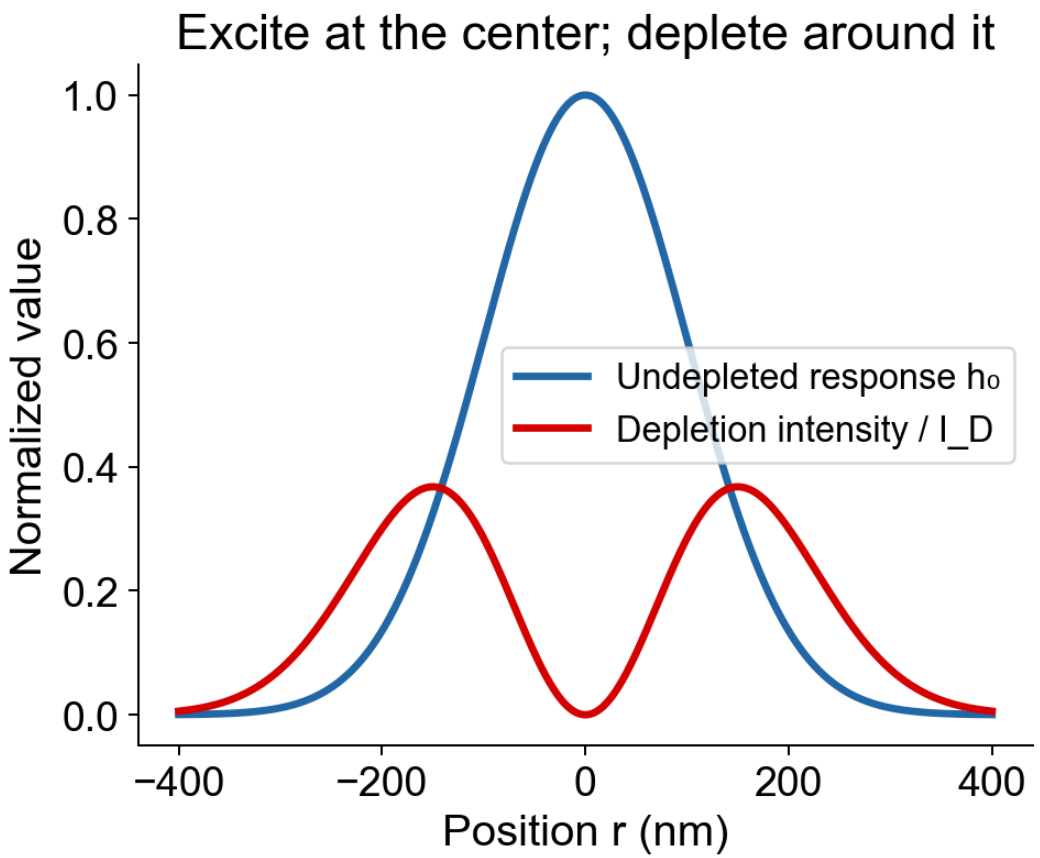
With STED on, stimulated decay is nine times as fast as spontaneous decay.

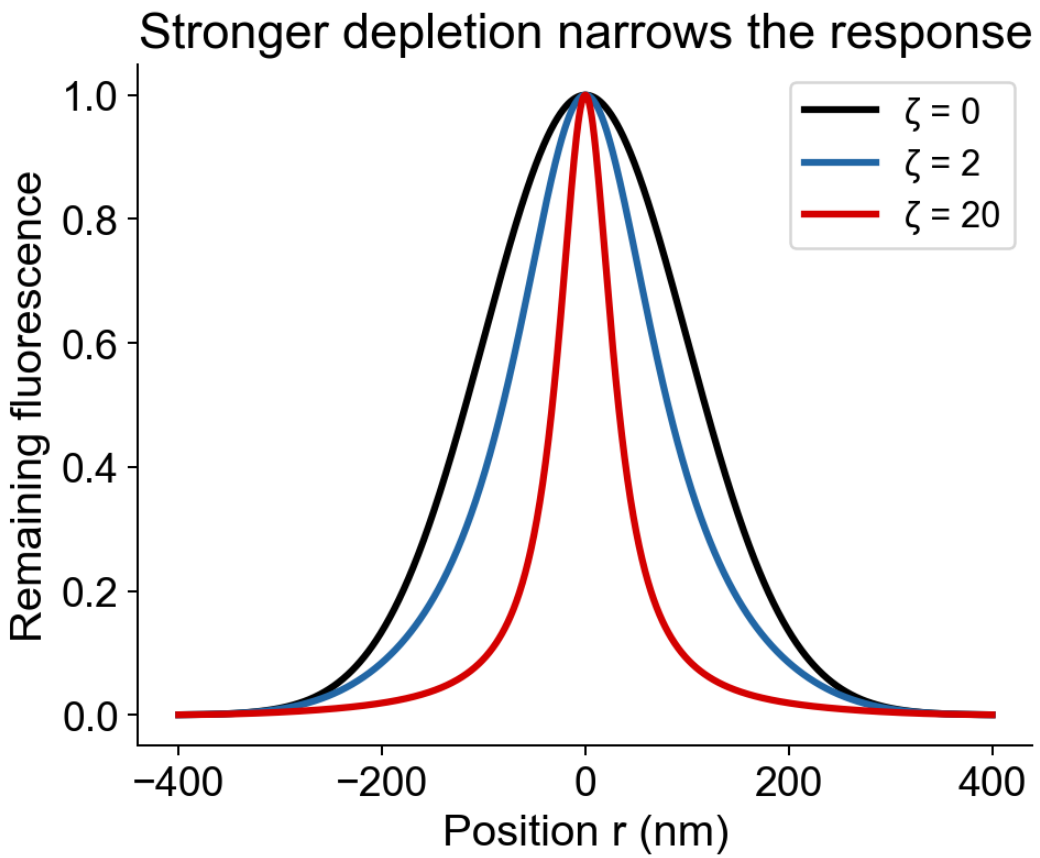
Expected outcome	STED off	STED on
Spontaneous photons emitted	100	10
Stimulated photons added to beam	0	90
Detected fluorescence (10% efficiency)	10	1

Spontaneous fraction = 1 / (1 + 9) = 0.10

100 radiative decays still occur, but only 10 use the spontaneous channel.
The other 90 add photons at the rejected depletion wavelength.

Place an intensity zero where fluorescence should survive





The depletion beam remains diffraction limited.
The surviving fluorescence region becomes narrower.

CW model: $I(r) = I_D(r/w)^2 \exp[-(r/w)^2]$, $\zeta = I_D/I_{sat}$, $s_0 = 100$ nm, $w = 150$ nm.

Scan the surviving fluorescence region across the specimen

At each scan position, the detector counts fluorescence that survives depletion.

1. Focus excitation onto one region.

2. Align the depletion zero with the excitation center.

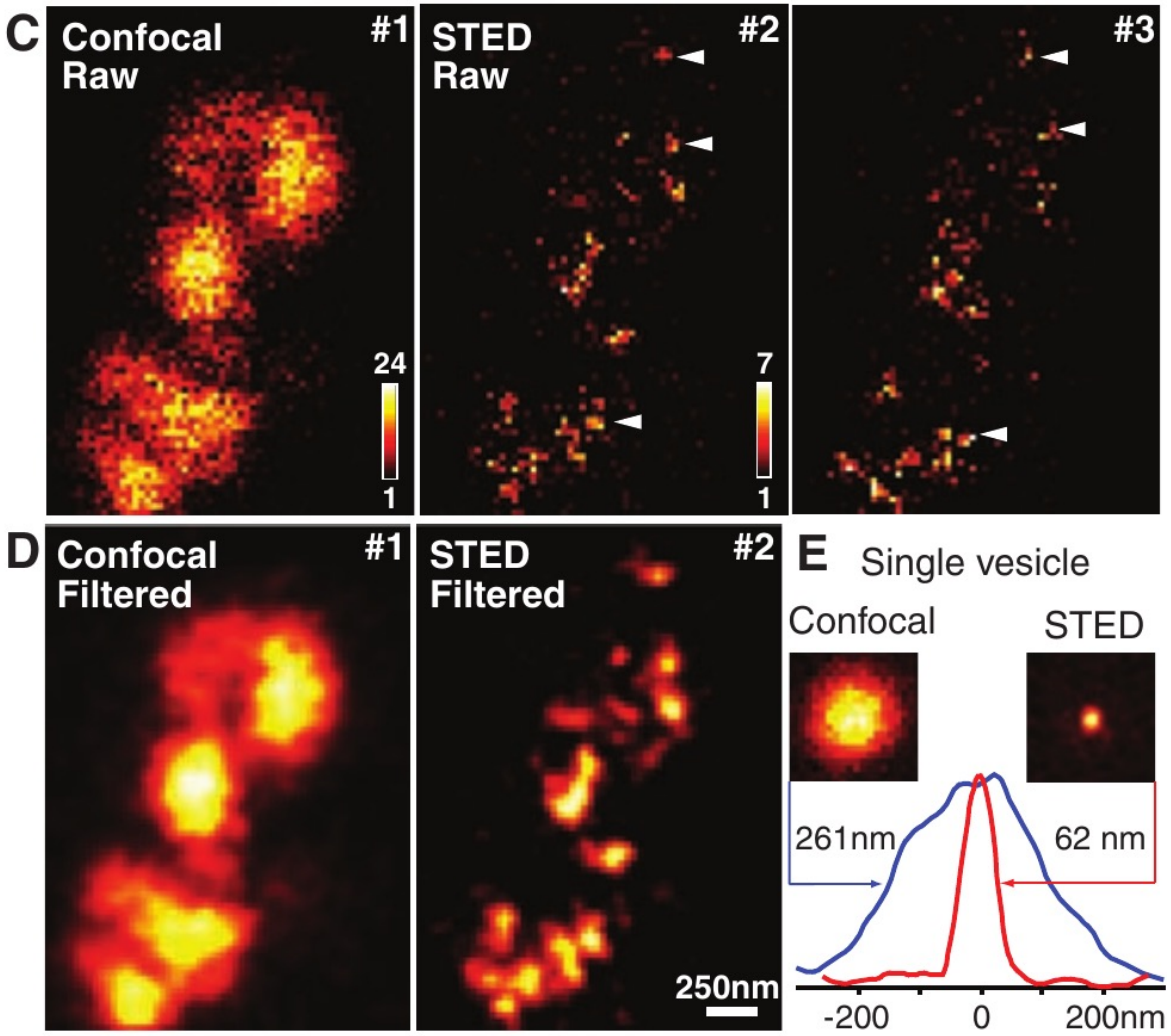
3. Count the remaining spontaneous fluorescence.

4. Move both beams together and repeat.

The scan coordinates locate each measurement.
The narrow effective response separates nearby structures.

STED commonly uses a scanning detector; camera pixel size is not the defining limit here.

STED separates structures that a confocal image merges



Fluorescent vesicles in
a living neuron.

Small-field imaging:
28 frames per second.

Stationary-vesicle profiles:
261 nm confocal
62 nm STED (FWHM).

Westphal et al., Science (2008), Fig. 1C–E. Live-neuron vesicles; scale bar 250 nm in panel D.

STED gains detail at a physical cost

Increasing depletion intensity improves confinement only if the experiment remains well controlled.

High intensity can increase bleaching and perturb the specimen.

Aberrations change the doughnut and the effective response.

Scanning speed, photon count and motion set the useful time resolution.

STORM and STED solve the overlap problem differently

	STORM	STED
What we control	Which labels are ON in time	Where fluorescence survives
Raw data	Many sparse camera frames	Counts at scan positions
Inference	Fit emitters; combine localizations	Assemble a scanned image
Useful strengths	Dense molecular maps; flexible labeling	Direct imaging; repeated scanning
Key constraints	Switching, coverage, drift, acquisition time	Intensity, zero quality, scan speed

Choose the method from the biological quantity, required time resolution and available labels.

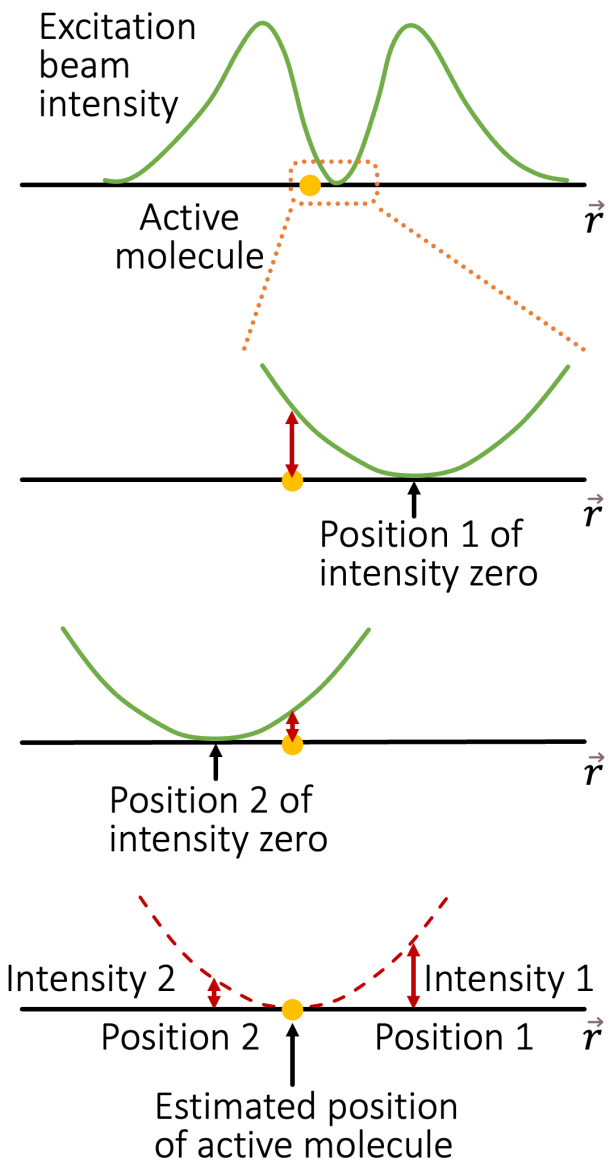
These are typical implementations; hybrid and specialized methods extend both families.

MINFLUX



from abberior: https://abberior.rocks/superresolution-confocal-systems/minflux/

MINFLUX



CROSS-SECTION OF DOUGHNUT-SHAPED EXCITATION BEAM

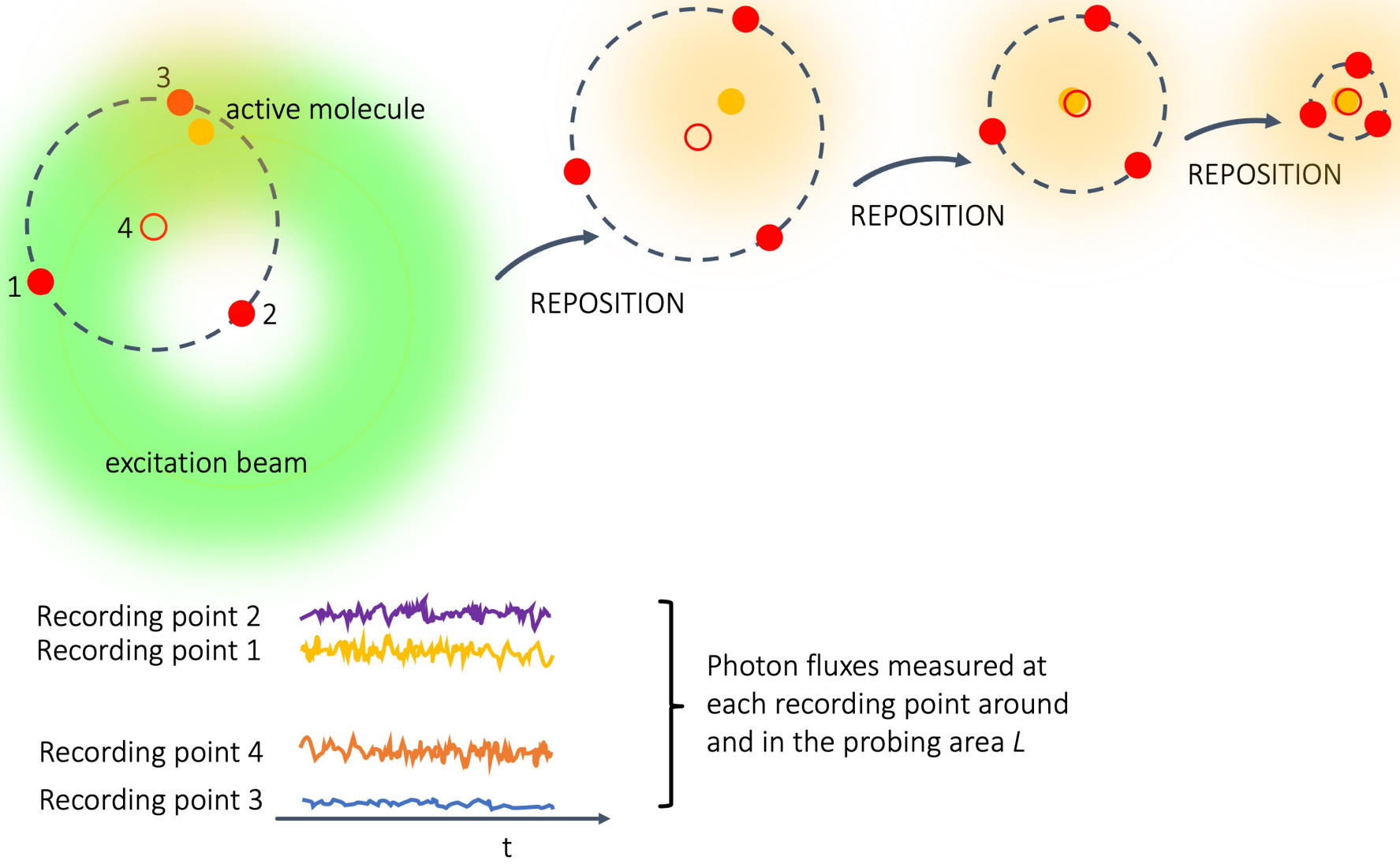
FIRST INTENSITY MEASUREMENT

SECOND INTENSITY MEASUREMENT

LOCALIZATION VIA QUADRATIC FIT

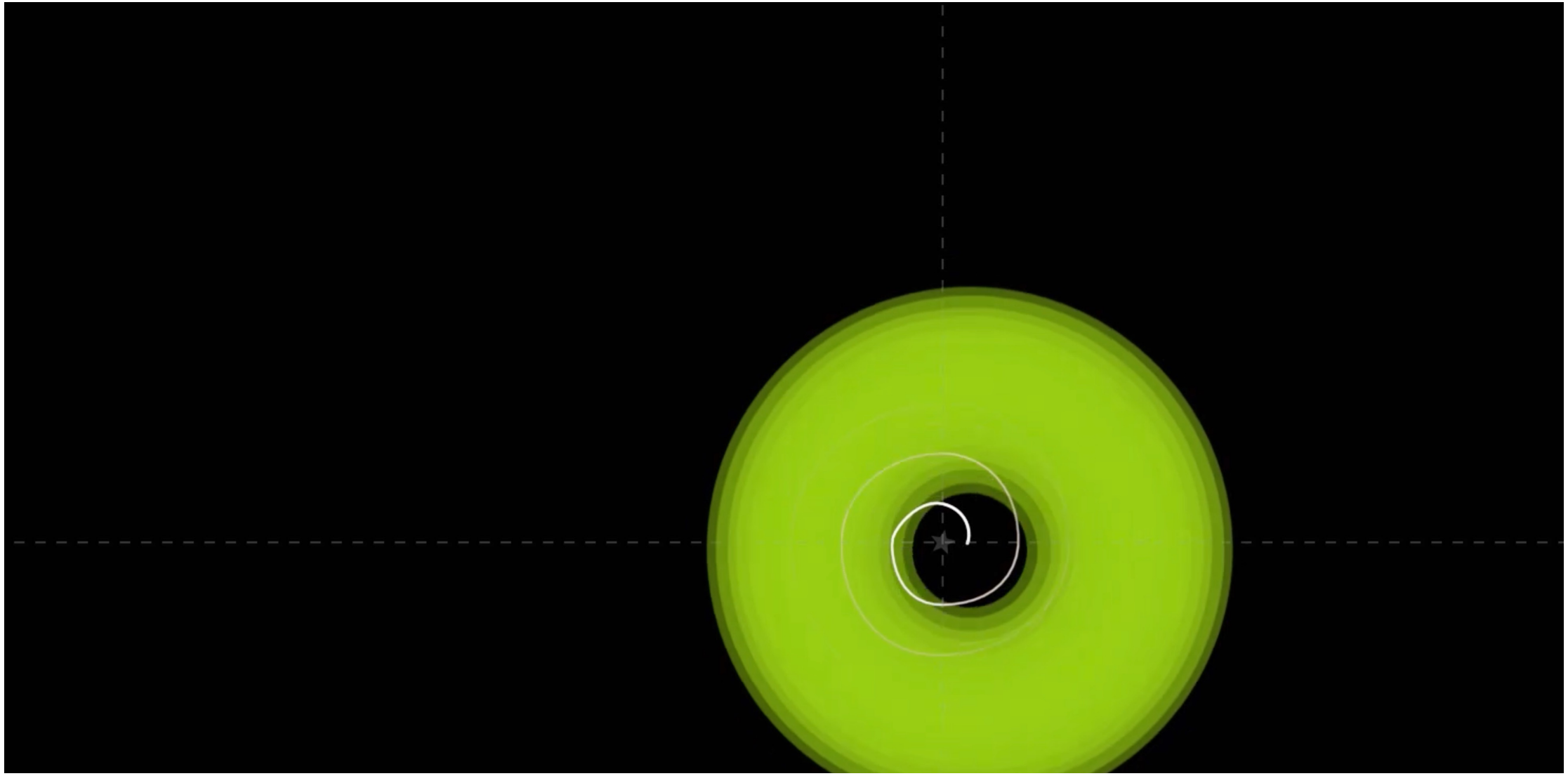
from Wiki: https://en.wikipedia.org/wiki/Minflux#/media/File:MINFLUX_principle_(described_in_one_dimension).png

MINFLUX



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MINFLUX



from abberior: https://abberior.rocks/superresolution-confocal-systems/minflux/

Shrink the search region while keeping the molecule inside

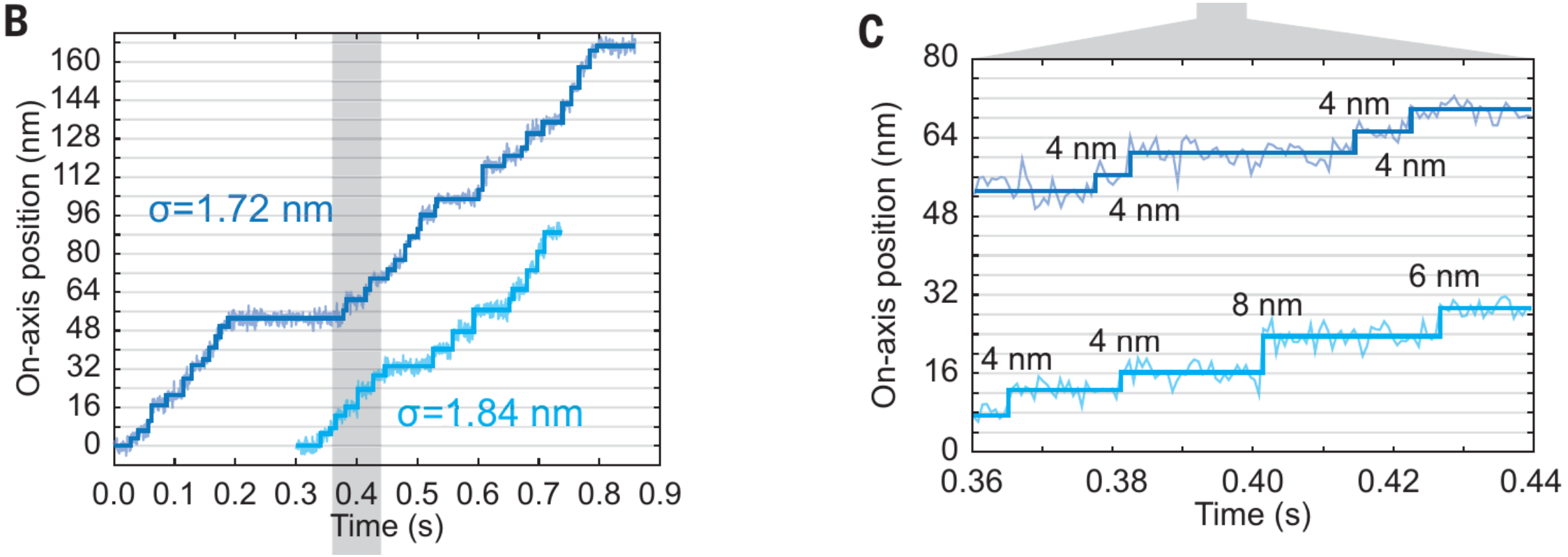
Real MINFLUX uses several probe positions and adapts their spacing.

1. Find an isolated emitter coarsely.
2. Probe around that estimated location.
3. Update the position from the photon counts.
4. Reduce the probe spacing and repeat, or follow a moving emitter.

A narrow probe region is useful only when the emitter remains inside it and the pattern is calibrated.

Balzarotti et al. (2017); subsequent MINFLUX developments extend tracking, 3D operation and multiplexed imaging.

MINFLUX resolves substeps of kinesin



~1-ms positions with ~1.7-nm precision.
The enlarged traces reveal ~4-nm substeps.

Wolff et al., Science (2023), Fig. 2B–C. Stalk-labeled kinesin N356C in vitro, 10 μM ATP.

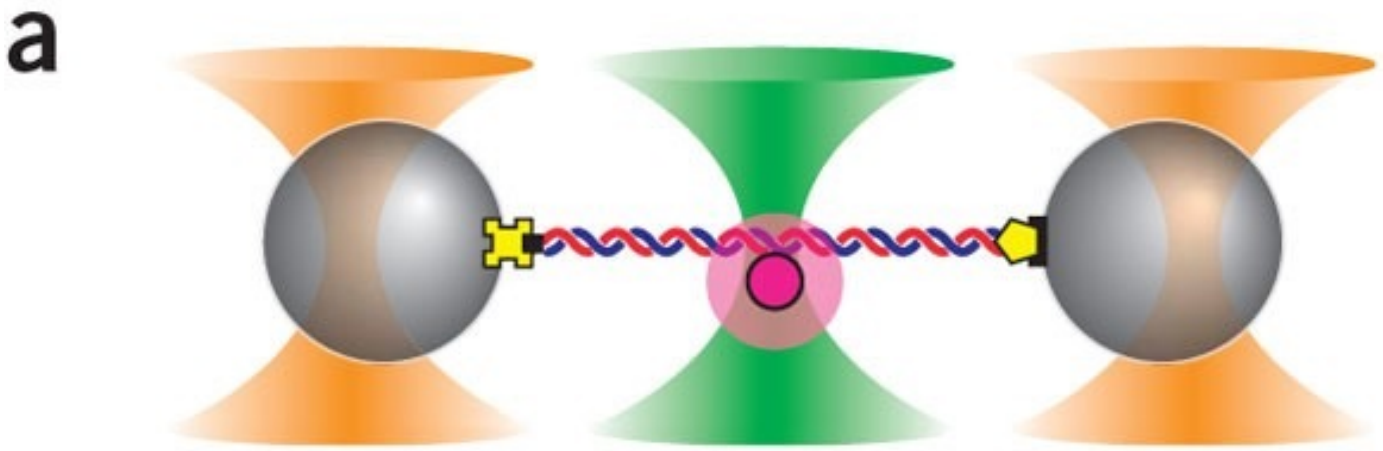
Different methods make different measurements possible

Biological aim	Useful starting point	What to check first
Map a dense fixed structure	STORM / PALM / DNA-PAINT	Label density, switching, linkage, drift
Repeatedly image a small live region	STED	Label tolerance, scan speed, photon budget
Follow one molecule or motor rapidly	MINFLUX or conventional tracking	Isolation, capture range, motion, background

Important: resolution, localization precision, field of view and time resolution.

Optical tweezers: force from a displaced bead

Focused light traps a dielectric bead.
Near equilibrium, the restoring force is approximately linear.



$$F_{\text{trap}} = -\kappa_t x_b$$

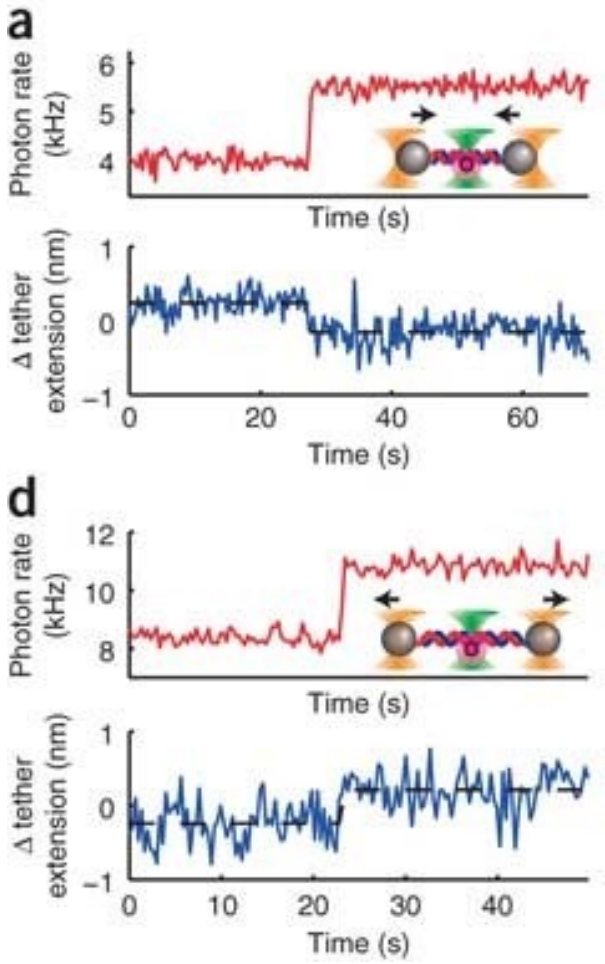
$$U_{\text{trap}} = \frac{1}{2}\kappa_t x_b^2$$

$$\kappa_t = 0.10 \text{ pN/nm}, \quad |x_b| = 30 \text{ nm} \implies |F| = 3 \text{ pN}$$

The bead reports force through a calibrated stiffness.
Fluorescence reports the labeled molecule.

Schematic: Comstock, Ha & Chemla, Nature Methods (2011), Fig. 1a. Numerical example is illustrative.

One binding event, two measured signals



A fluorescent 9-nucleotide DNA strand binds to a complementary gap.

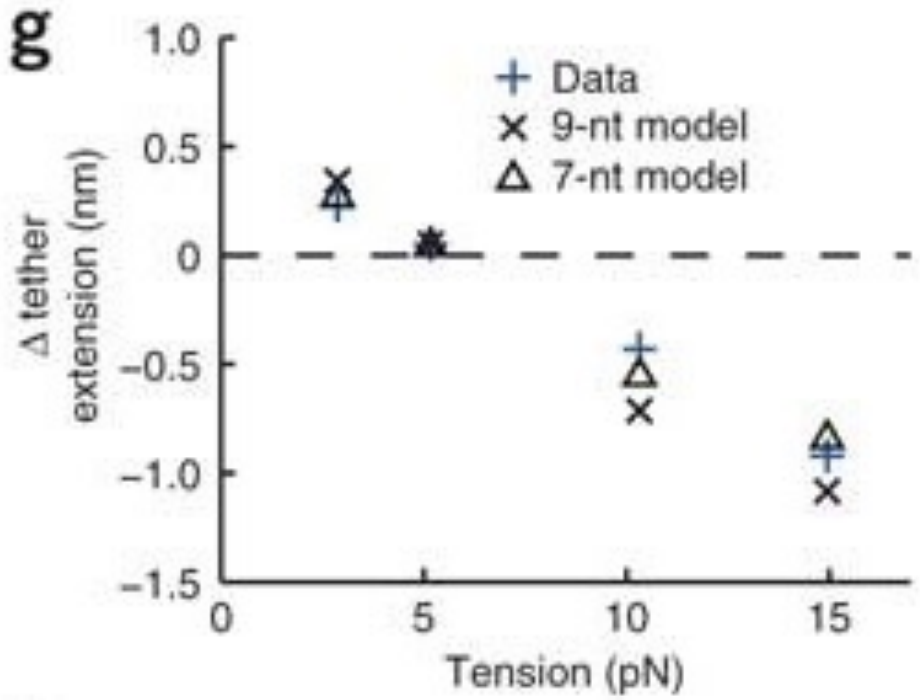
At 10 pN, binding raises fluorescence and shortens the tether.

At 3 pN, binding raises fluorescence and lengthens the tether.

The same chemical event can move the mechanical signal in opposite directions.

Comstock, Ha & Chemla (2011), Fig. 4a,d. Red: photon rate. Blue: tether extension change.

A mechanical blind spot near 5.5 pN



Binding converts a flexible single-stranded segment into a stiffer duplex.

Their projected extensions depend differently on force.

At the crossing, binding changes molecular state but barely changes tether extension.

$$\Delta X(F) \simeq A(F)\, n\, [x_{\mathrm{ds}}(F) - x_{\mathrm{ss}}(F)]$$

A second reporter can reveal a transition that extension alone cannot identify.

Comstock, Ha & Chemla (2011), Fig. 4g. A(F) is a positive mechanical transmission factor. The crossing depends on the construct.

What could tweezers plus MINFLUX measure?

Measurement	What it reports	A question it helps answer
Calibrated bead displacement	Force and tether extension	How does applied load change the event?
MINFLUX localization	Position of a fluorescent label	Where does the labeled domain move?
A synchronized molecular reference	Relative displacement	How much motion comes from translation of the whole construct?

Compare motion under known load with motion of a specific labeled part.

Proposed measurement. Combined tweezers and fluorescence are established; the MINFLUX integration requires its own validation.

Summary

We learn position by combining photon statistics with a physical model of the measurement.

More independent photons narrow uncertainty in a fixed center.

Confocal and light-sheet microscopy reduce background from other planes.

STORM separates emitters in time by controlling bright and dark states.

STED suppresses spontaneous fluorescence around an intensity zero.

MINFLUX reports label position. Tweezers add force and a mechanical coordinate.

Interpret optical and mechanical coordinates through the labels, handles and measurement model.

End of the main lecture. Optional derivations and supporting material follow.